

Unit 9

THE PERIODIC TABLE: CHEMICAL PERIODICITY

79 items · 375 marks

Syllabus sections in this unit

- 9.1 Periodicity of physical properties of the elements in Period 3
- 9.2 Periodicity of chemical properties of the elements in Period 3
- 9.3 Chemical periodicity of other elements

Unit 9: The Periodic Table: chemical periodicity**1. M/J 26/P22/Q1/c,d**

1 Oxygen reacts with many elements.

(c) Li_2O reacts with water to produce a solution with pH greater than 10.

(i) State the name of a Period 3 oxide that behaves in the same way as Li_2O when added to water.

[1]

(ii) Construct an equation for the reaction of Li_2O with water.

[1]

(iii) Identify **all** the Period 3 chlorides that are solid at room temperature and that react with water to produce an acidic solution.

[2]

(d) LiNO_3 decomposes on heating to produce Li_2O , O_2 and a brown gas.

Identify the brown gas.

[1]

2. M/J 26/P23/Q1/a

1 The chlorides of Period 3 elements show different physical and chemical properties.

(a) (i) Complete Table 1.1.

Table 1.1

element	formula of chloride	type of bonding in the chloride	type of reaction, if any, of the chloride with cold water
sodium			
silicon			

[3]

(ii) Write an equation for the reaction of PCl_5 with an excess of cold water.

[1]

3. M/J 26/P24/Q3/a,b

3 (a) (i) State the trend in atomic radius of the elements across Period 3.

[1]

(ii) State the formula of the oxide formed when sodium reacts with oxygen.

[1]

(iii) Write an equation for the reaction of MgO with water.

[1]

(b) Table 3.1 shows some of the physical properties of Period 3 chlorides.

Table 3.1

	melting point /°C	boiling point /°C	type of bonding	type of structure
NaCl	801	1413		
SiCl₄	−69	58		

(i) Complete Table 3.1 using the abbreviations shown.

M = metallic

C = covalent

I = ionic

S = simple

G = giant

[2]

(ii) State the formula of the phosphorus-containing product formed when PCl₅ reacts with an excess of water. Suggest the pH of the resulting solution.

formula =

pH of solution =

[2]

4. O/N 25/P21/Q2/a,b,c

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to Cl.

The ionic radius of Si is **not** shown.

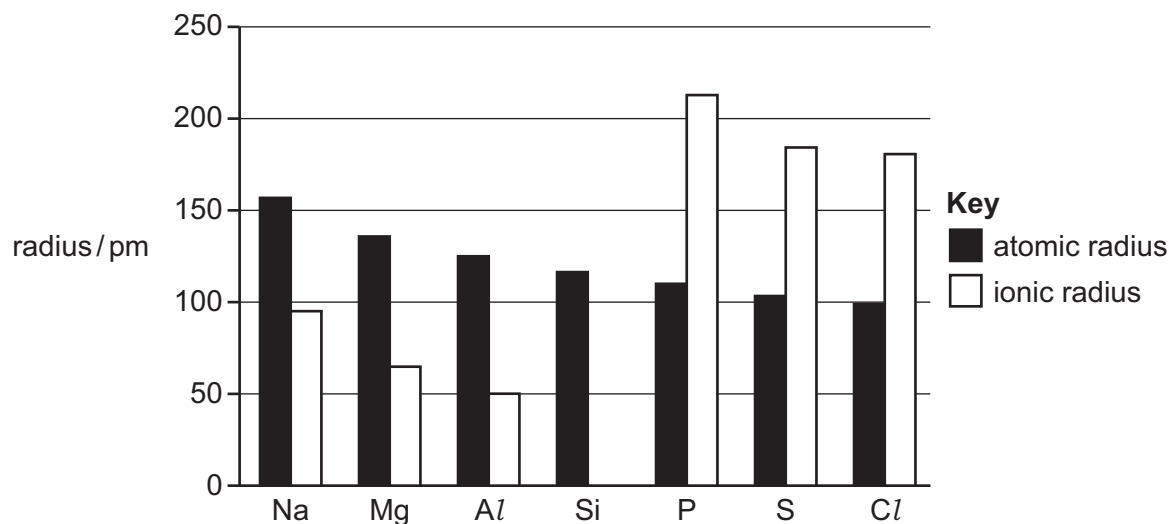


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to Cl.

[2]

(ii) Explain why there is a large difference in the ionic radii of Al and P.

[2]

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₃
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1.

[2]

(ii) State why there is no data given in row B for Al₂O₃ and SiO₂.

[1]

(c) (i) Write an equation for the reaction of Na₂O with dilute hydrochloric acid.

[1]

(ii) Construct an equation for the reaction of Al₂O₃ with a base to form NaAlO₂.

[1]

5. O/N 25/P22/Q2/a,b,c,d

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) (i) Explain why the elements Na to Al are good electrical conductors.

[1]

(ii) Explain why the elements P, S and Cl do not conduct electricity.

[1]

(b) Fig. 2.1 shows the variation in melting point of the Period 3 elements Si to Cl.

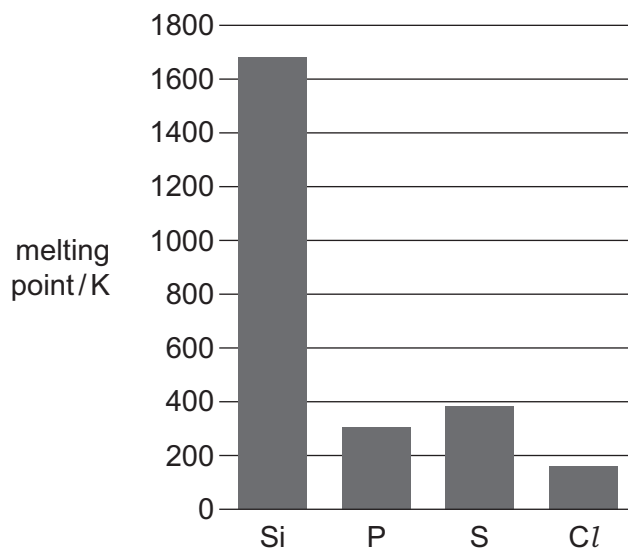


Fig. 2.1

The Period 3 elements Si to Cl are all non-metals.

Explain why there is a large difference between the melting point of Si and the melting points of P, S and Cl.

[2]

(c) Table 2.1 gives some information about some Period 3 chlorides.

Row B refers to the pH of the solution that forms when the Period 3 chloride is added to water.

Table 2.1

	formula of Period 3 chloride	NaCl	MgCl ₂	AlCl ₃	SiCl ₄	PCl ₅
A	oxidation number of element bonded to Cl					
B	pH of solution		6.5			
C	bonding			ionic		
D	structure			giant		

Complete Table 2.1.

You may use the following abbreviations.

I = ionic, C = covalent, M = metallic
G = giant, S = simple

[4]

(d) (i) Write an equation for the formation of $AlCl_3$ from its elements.

[1]

(ii) Write an equation for the formation of H_3PO_4 from PCl_5 .

[1]

6. O/N 25/P23/Q2/a,b,c

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to Cl.

The ionic radius of Si is **not** shown.

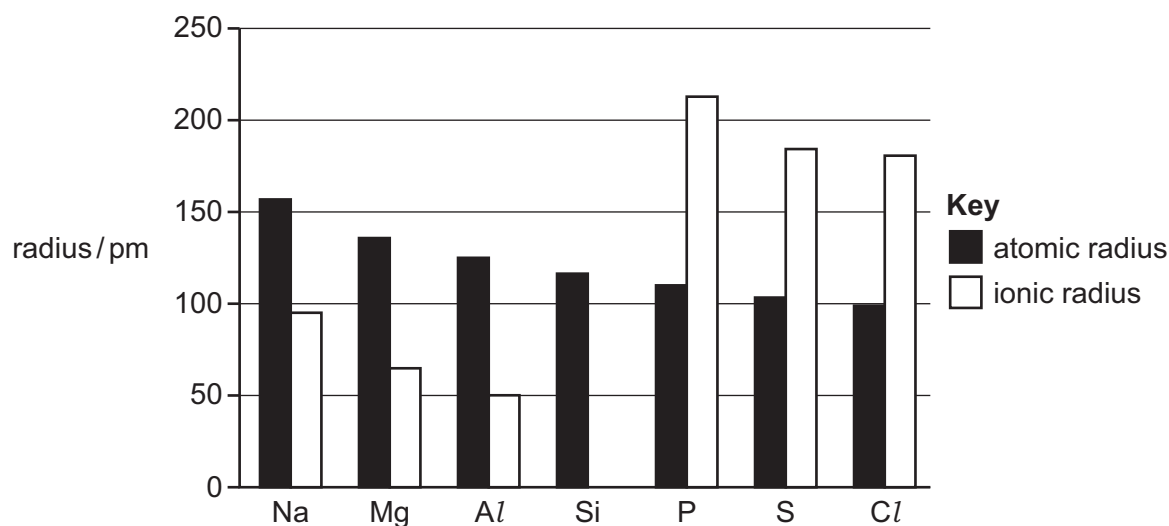


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to Cl.

[2]

(ii) Explain why there is a large difference in the ionic radii of Al and P.

[2]

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1.

[2]

(ii) State why there is no data given in row B for Al_2O_3 and SiO_2 .

[1]

(c) (i) Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

[1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form $NaAlO_2$.

[1]

7. O/N 25/P24/Q1/a(ii),d

1 The Group 17 elements are oxidising agents.

(ii) Write an equation to show the reaction in which Cl_2 oxidises aluminium metal.

[1]

(d) Table 1.3 gives some information about $MgCl_2$ and $SiCl_4$.

Table 1.3

	$MgCl_2$	$SiCl_4$
electrical conductivity when liquid	conducts	does not conduct
observation when added to water	dissolves	vigorous reaction

(i) Explain the difference between the electrical conductivity of liquid $MgCl_2$ and of liquid $SiCl_4$. Refer to bonding and relevant particles in your answer.

[2]

(ii) Suggest the pH of the solutions that form when each chloride is added to water.

$MgCl_2$

$SiCl_4$

[2]

8. O/N 25/P24/Q2/c(iii)

2 Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

(b) A 5.00 dm^3 sealed flask contains 0.400 mol of CO(g) and 0.800 mol of $\text{H}_2\text{(g)}$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.



The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$$

(c) P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH .



P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .

(iii) Write an equation to show the reaction of P_4O_{10} with an excess of water.

[1]

9. M/J 25/P21/Q1/b,c

(b) Separate samples of sodium and sodium oxide are each added to an excess of cold water.

(i) Write an equation for the reaction of sodium with cold water.

[1]

(ii) Write an equation for the reaction of sodium oxide with cold water.

[1]

(iii) Complete Table 1.1.

Do **not** refer to temperature changes when considering observations for these reactions.

Table 1.1

	sodium	sodium oxide
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		

[2]

(c) State the oxidation number of the Period 3 elements bonded to Cl in NaCl and PCl_5 . Explain the difference in the oxidation number.

[2]

10. M/J 25/P22/Q1/b,c

(b) Separate samples of phosphorus(V) chloride and silicon(IV) chloride are each added to an excess of cold water.

(i) Write an equation for each reaction.

[2]

(ii) Describe the appearance of phosphorus(V) chloride and silicon(IV) chloride at room temperature.

phosphorus(V) chloride

silicon(IV) chloride

[2]

(iii) Compare the appearance of the mixtures produced when each reaction is complete.

[2]

(c) Some oxides are amphoteric.

(i) Describe what is meant by amphoteric.

[1]

(ii) Identify the formula of a Period 3 oxide that is amphoteric.

[1]

11. M/J 25/P23/Q2/a,b,c(i),e

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

- (a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water.
State the formula of the oxide.

[1]

- (b) P_4O_{10} is added to an excess of aqueous NaOH.
Write an equation to describe the reaction.

[1]

- (c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point/ $^{\circ}\text{C}$
SO_2	-73
H_2O	0
SO_3	17
SiO_2	1610
Na_2O	1132
MgO	2852
Al_2O_3	2072

- (i) Identify the oxide from Table 2.1 that contains the element with the highest oxidation number.

[1]

- (e) Al_2O_3 is a white amphoteric compound.

- (i) State the formula of the aluminium-containing species produced when Al_2O_3 reacts with NaOH(aq) .

[1]

- (ii) State the formula of the aluminium-containing salt produced when Al_2O_3 reacts with $\text{H}_2\text{SO}_4\text{(aq)}$.

[1]

12. M/J 25/P24/Q1/b,d

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(b) The relative electrical conductivities of the Period 3 elements are shown in Fig. 1.1.

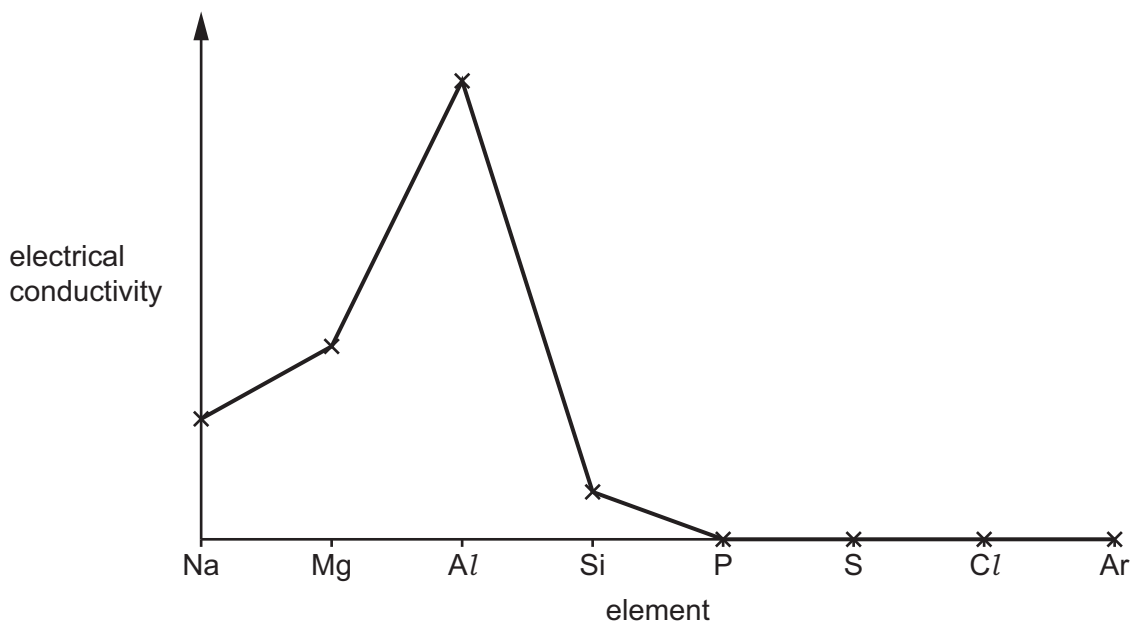


Fig. 1.1

Explain why there is an increase in conductivity from Na to Al and why P, S and Cl are non-conductors of electricity.

[2]

(d) Most Period 3 elements react with oxygen to form oxides.

(i) Write an equation for the reaction of phosphorus with oxygen.

[1]

(ii) Write an equation for the reaction of aluminium oxide with an excess of aqueous sodium hydroxide.

[1]

13. M/J 25/P24/Q2/b(i,iii)

(b) SCl_2 is a red liquid that reacts with water.



(i) Describe **two** observations, other than temperature change, that are made when an excess of water is added to SCl_2 .

[2]

(iii) State the general name to describe the type of reaction that occurs when a substance is broken down by water.

[1]

14. MAR 25/P22/Q1/b(i,iii,iv,v)

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

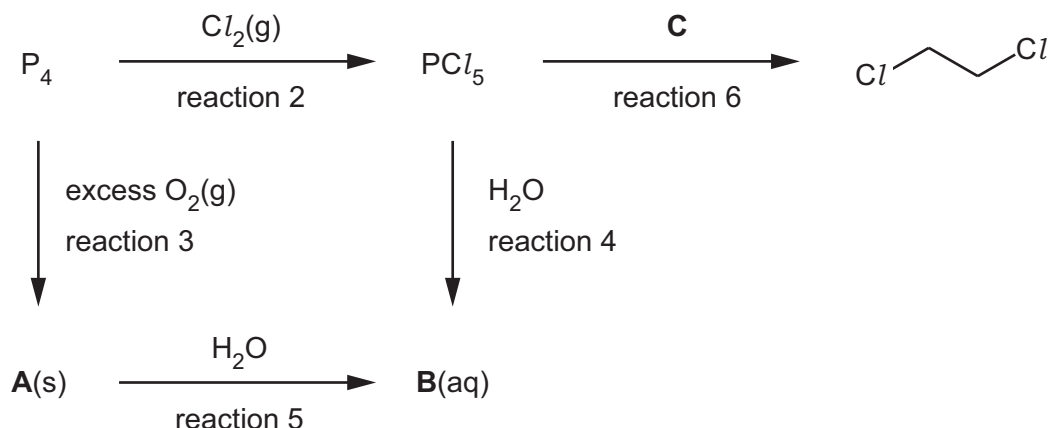


Fig. 1.1

(i) Suggest what is observed in reaction 2.

[1]

(iii) Write an equation for the formation of **A** in reaction 3.

[1]

(iv) Name **B**, which is formed in both reactions 4 and 5.

[1]

(v) Draw the structure of **C**, used in reaction 6.

[1]

15. O/N 24/P21/Q3/e(i,ii),f

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

[1]

(ii) Describe what is observed when a small sample of $SiCl_4$ is added to water.

[1]

(f) Tin forms an amphoteric oxide, SnO_2 .

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

[1]

16. O/N 24/P22/Q2/a(i,iii),b(i)

2 Oxygen is a Group 16 element.

(a) (i) Write equations for the following reactions.

- sodium and oxygen
- sulfur and oxygen

[2]

(iii) The maximum oxidation state of the Period 3 elements in their oxides varies across the period.

State and explain the variation.

[2]

(b) H_2O reacts with both inorganic and organic compounds.

(i) Complete Table 2.1 to give details of the reactions of some Period 3 oxides with H_2O .

Table 2.1

Period 3 oxide	product of reaction with H_2O	pH of solution formed
	$\text{Mg}(\text{OH})_2$	
P_4O_{10}		

[2]

17. O/N 24/P23/Q3/e(i,ii),f

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of SiCl_4 from its constituent elements.

[1]

(ii) Describe what is observed when a small sample of SiCl_4 is added to water.

[1]

(f) Tin forms an amphoteric oxide, SnO_2 .

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

[1]

18. M/J 24/P21/Q2/a

- 2 (a) Sulfur chloride, SCl_2 , is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

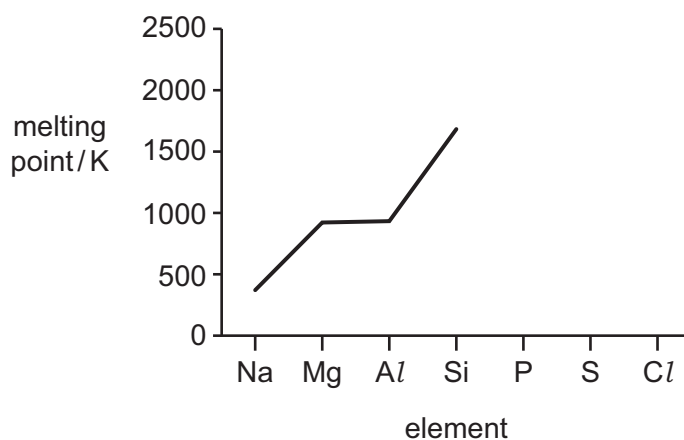
[1]

(ii) Name a chloride of a different Period 3 element that is also a liquid at room temperature and produces misty fumes when added to water.

[1]

19. M/J 24/P22/Q1/d,e,f

- (d) Fig. 1.1 shows the variation in melting point of some Period 3 elements in their standard states at room temperature and pressure.

**Fig. 1.1**

- (i) Explain why Si has a high melting point.

[1]

(ii) Complete Fig. 1.1 to show the variation in the melting points of the elements P, S and Cl.

[2]

- (e) Two Period 3 elements react with an excess of oxygen at room pressure.

(i) Complete Table 1.2.

Table 1.2

1	2	3
Period 3 element	state of oxide at room temperature and pressure	approximate pH of solution made when oxide is added to water
Na		
S		

[2]

- (ii) The solutions made in column 3 of Table 1.2 are mixed together. Name the type of reaction that occurs.

[1]

(iii) Write an equation to describe the reaction between P_4O_{10} and an excess of water.

[1]

(f) Aluminium hydroxide is amphoteric.

(i) Explain what is meant by amphoteric.

[1]

(ii) Write an equation to describe the reaction that occurs when aluminium hydroxide, $Al(OH)_3$, reacts with $NaOH(aq)$.

[1]

20. M/J 24/P23/Q1/b,c

(b) (i) A sample of Al reacts with an excess of Cl_2 .

State the oxidation number of Al in the product of the reaction.

oxidation number of Al

[1]

(ii) State what determines the maximum oxidation number of the Period 3 elements in their oxides.

[1]

(c) Separate samples of aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , react with an excess of $NaOH(aq)$ at room temperature.

(i) Give the state of Al_2O_3 and P_4O_{10} at room temperature.

Al_2O_3

P_4O_{10}

[1]

(ii) Write an equation for the reaction of each oxide with an excess of $NaOH(aq)$ at room temperature.

Al_2O_3 +

P_4O_{10} +

[2]

21. MAR 24/P22/Q2/a,c(ii)

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

- (a)** Complete Table 2.1.

[3]

- (c)** An excess of Cl_2 reacts with phosphorus to form PCl_5 .

- (ii)** PCl_5 reacts with **J** to form H_3PO_4 .

Identify **J** and state the type of reaction.

J type of reaction

[2]

22. O/N 23/P21/Q1/d(ii)

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (d) POC_l₃ shows similar chemical properties to PC_l₅.

POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (ii) Construct an equation for the reaction of POC_l₃ with water.



[1]

23. O/N 23/P22/Q1/a

- 1 The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

- (a) (i) Describe the variation in atomic radius from silicon to sulfur.

[1]

- (ii) The melting point of silicon is 1410°C. The melting point of sulfur is 113°C.

Explain this difference.

[3]

24. O/N 23/P22/Q3/a(i)

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4 .

(i) Write an equation for each step.

step 1

step 2

[2]

25. O/N 23/P23/Q1/d(ii)

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^-

(d) POCl_3 shows similar chemical properties to PCl_5 .

POCl_3 has a melting point of 1°C and a boiling point of 106°C .

POCl_3 reacts vigorously with water, forming misty fumes and an acidic solution.

(ii) Construct an equation for the reaction of POCl_3 with water.



[1]

26. M/J 23/P21/Q1/f

- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(f) TeF_x reacts with water to form tellurium hydroxide and HF. The oxidation number of tellurium does **not** change during this reaction.

(i) Construct an equation for the reaction of TeF_x with water.

[1]

(ii) Name the type of reaction that occurs when TeF_x reacts with water.

[1]

27. M/J 23/P21/Q3/a,b

- 3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance

structure and bonding

[2]

(b) Samples of magnesium chloride and phosphorus(V) chloride are added to separate beakers of cold water.

Complete Table 3.1. Ignore temperature changes when considering observations for these reactions.

Table 3.1

	magnesium chloride	phosphorus(V) chloride
appearance at room temperature		
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		
pH of final solution		

[4]

28. M/J 23/P22/Q1/a(iii)

- 1** The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point/K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (iii)** Explain the difference in the melting points of phosphorus and sulfur in terms of structure and bonding.

[2]

29. M/J 23/P22/Q2/a,b,c

2 Chlorine is a reactive element. It forms many compounds.

- (a) (i)** Complete Table 2.1 to show the maximum oxidation number of the elements Na to P in their chlorides.

Table 2.1

element	Na	Mg	Al	Si	P
maximum oxidation number					

[1]

- (ii)** State what determines the maximum oxidation number of elements in Period 3.

[1]

(b) An excess of cold water is added to the chloride of silicon.

- (i)** Write an equation for the reaction between an excess of cold water and the chloride of silicon.

[1]

- (ii)** Suggest the pH of the solution produced in **(b)(i)**.

[1]

(c) An excess of cold water is added to the chloride of phosphorus.

- (i)** Write an equation for the reaction between an excess of cold water and the chloride of phosphorus.

[1]

- (ii)** Suggest the pH of the solution produced in **(c)(i)**.

[1]

30. M/J 23/P23/Q2/a,c

- 2 (a)** The reaction of pure aluminium is only observed if the aluminium oxide layer is removed first. When pure aluminium is added to cold water, bubbles of gas are seen.

(i) State **one** property of aluminium oxide that explains why an aluminium object does **not** react with cold water until the aluminium oxide layer is removed.

[1]

(ii) Write an equation, with state symbols, for the reaction of aluminium oxide with an excess of NaOH(aq).

[2]

(iii) Name **one** other Period 3 element that also produces bubbles of gas when added to cold water.

[1]

(c) Aluminium chloride and phosphorus chloride are both white solids.

(i) State the maximum oxidation number of aluminium and of phosphorus in these solid chloride salts.

maximum oxidation number of aluminium

maximum oxidation number of phosphorus

[1]

(ii) State why the maximum oxidation number of aluminium is different from that of phosphorus.

[1]

(iii) Write an equation for the reaction of solid phosphorus chloride and excess water.

[1]

(iv) Name the type of reaction that occurs when aluminium chloride is added to water.

[1]

(v) Explain why the solution produced after aluminium chloride is added to water has a pH of 1–2.

[1]

31. MAR 23/P22/Q1/d(i,ii)

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy / kJ mol ⁻¹	—	7300	2350	3390	2260

(d) CO₂ and SO₂ are acidic gases.

(i) Write an equation for the reaction of SO₂ with H₂O.

[1]

(ii) Write an equation for the reaction of SO₂ with NaOH.

[1]

32. O/N 22/P21/Q3/a

- 3 Some of the common chlorides of Period 3 elements are shown in the list.



(a) From this list, identify:

(i) all the chlorides that have giant ionic structures in the solid state

[1]

(ii) all the chlorides that react vigorously with water to form strongly acidic solutions

[1]

(iii) the chloride that dissolves in water to form a neutral solution

[1]

(iv) the chloride formed from the **element** with the highest melting point.

[1]

33. O/N 22/P22/Q2/a,b

2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(a) Complete Table 2.1.

- Identify the bonding shown by each chloride under standard conditions. Use C = covalent, I = ionic, M = metallic.
- Identify the structure shown by each chloride under standard conditions. Use G = giant, S = simple.
- Deduce the oxidation state of the Period 3 element in each chloride.

[4]

(b) Write equations for the reactions of NaCl and PCl_5 with water. Include state symbols in both equations.



[3]

34. O/N 22/P23/Q3/a

3 Some of the common chlorides of Period 3 elements are shown in the list.



(a) From this list, identify:

(i) all the chlorides that have giant ionic structures in the solid state

[1]

(ii) all the chlorides that react vigorously with water to form strongly acidic solutions

[1]

(iii) the chloride that dissolves in water to form a neutral solution

[1]

(iv) the chloride formed from the **element** with the highest melting point.

[1]

35. M/J 22/P21/Q1/f(i)

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.
- (f)** Magnesium, Mg, burns in oxygen, O₂.
The activation energy, E_a , for this reaction is +148 kJ mol⁻¹.
- (i)** State **one** observation when magnesium burns in oxygen.
Do **not** refer to temperature changes in your answer.

[1]

36. M/J 22/P21/Q3/d(iii)

3 Liquids that contain molecules of **T** smell like lemons.

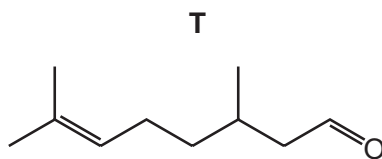


Fig. 3.1

(c) Fig. 3.2 shows two reactions of **T**.

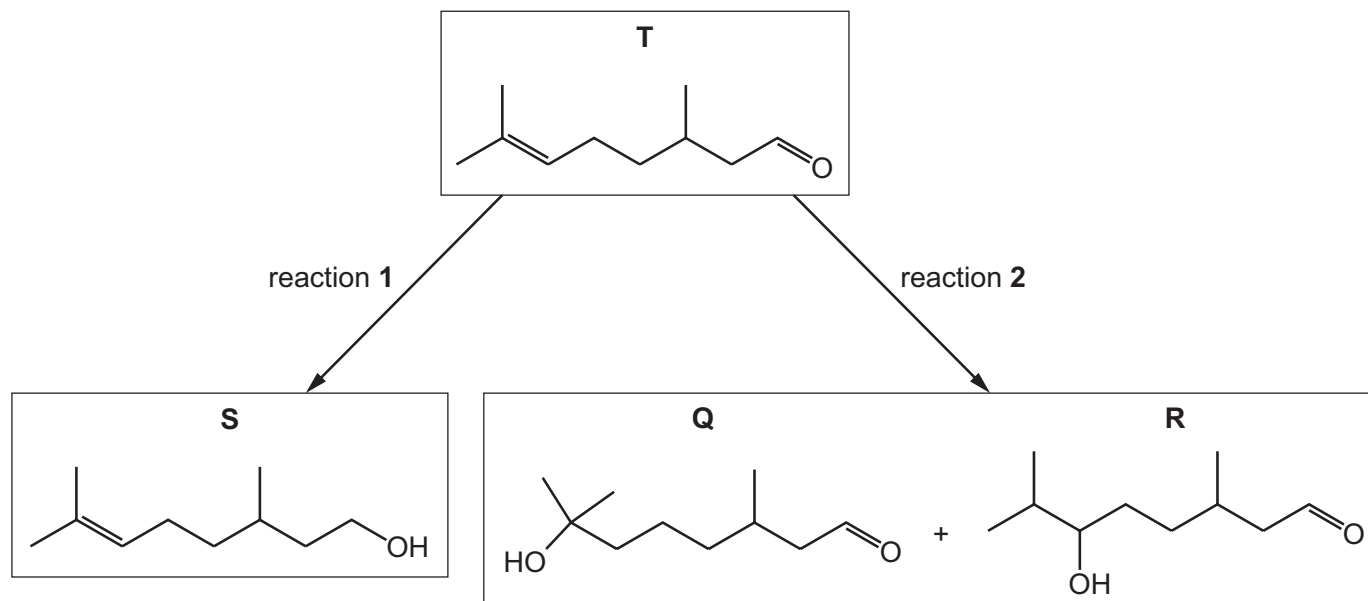


Fig. 3.2

(d) Separate samples of **Q** and **R** are added to separate test-tubes containing acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ and heated.

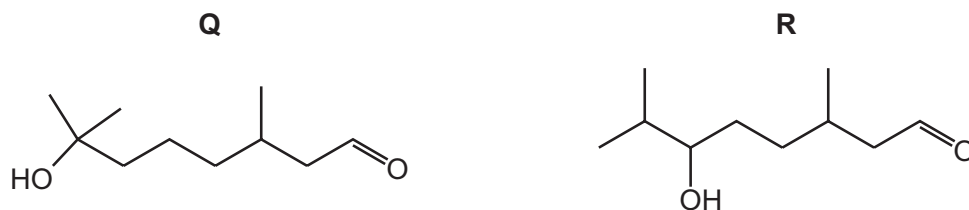


Fig. 3.3

(iii) Suggest why samples of **Q** and **R** must be dried before PCl_5 is added. Include a relevant equation to support your answer.

[2]

37. M/J 22/P23/Q2/a

2 (a) Period 3 elements and their compounds show trends in their physical properties.

(i) On Fig. 2.1 sketch a graph to show the melting points of the first five elements in Period 3.

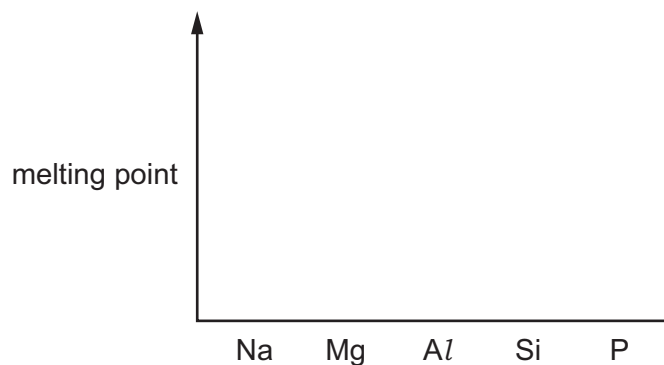


Fig. 2.1

[3]

(ii) Complete Table 2.1 with information for sodium chloride and phosphorus(V) chloride.

Table 2.1

	sodium chloride	phosphorus(V) chloride
state at room temperature		
name of change which occurs on addition of water		
pH of final solution		

[3]

38. MAR 22/P22/Q2/a(ii),b(i,ii),c(iv)

2 Some oxides of elements in Period 3 are shown.



(a) Na reacts with O_2 to form Na_2O . Na is the reducing agent in this reaction.

(ii) Write an equation for the reaction of Na_2O with water.

[1]

(b) Al_2O_3 is an amphoteric oxide found in bauxite.

(i) State what is meant by amphoteric.

[1]

(ii) Al_2O_3 is purified from bauxite in several steps. The first step involves heating Al_2O_3 with an excess of NaOH(aq) . A colourless solution forms.

Write an equation for this reaction.

[1]

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(iv) Write an equation for the reaction of P_4O_{10} with water.

[1]

39. MAR 22/P22/Q3/b(i)

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

(b) HCl is a product of several different reactions. Some of these are shown in Fig. 3.1.

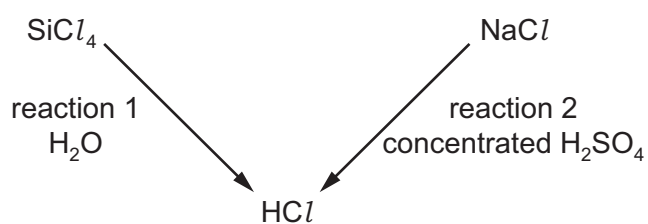


Fig. 3.1

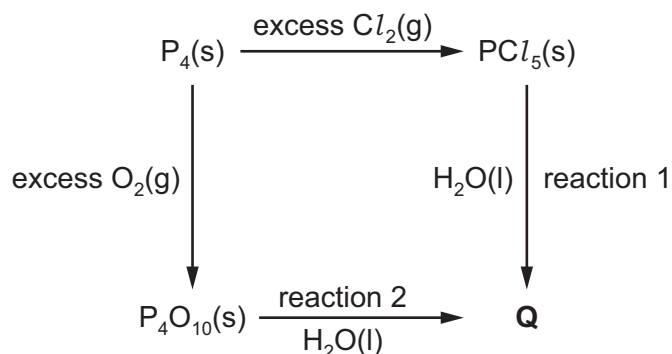
(i) Write an equation for reaction 1.

[1]

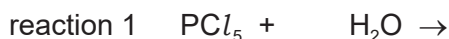
40. O/N 21/P21/Q3/b(ii)

3 Phosphorus is a reactive Period 3 element.

(b) Some reactions of $P_4(s)$ are shown in the reaction scheme.



(ii) Deduce the identity of **Q** and hence construct chemical equations for reactions 1 and 2.

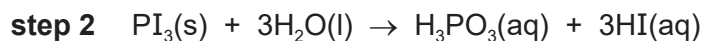
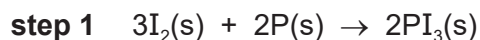


[2]

41. O/N 21/P22/Q1/f(ii)

1 Hydrogen iodide, HI, is a colourless gas at room temperature.

(f) In the laboratory, HI(aq) can be formed in a two-step process.



(ii) Name the type of reaction in **step 2**.

[1]

42. O/N 21/P22/Q2/a(i,iv),b,c,d(i,ii)

2 (a) Table 1 gives physical data for some of the Period 3 elements.

Table 1

atomic number, Z	11	12	13	14	15	16	17
bonding present in element	M						C
first ionisation energy / kJ mol ⁻¹	494	736	577	786	1060	1000	1260
maximum oxidation number							+7
anionic radius / nm	–	–	–	0.271	0.212	0.184	0.181

(i) Complete the row in the table labelled 'bonding present in element'.

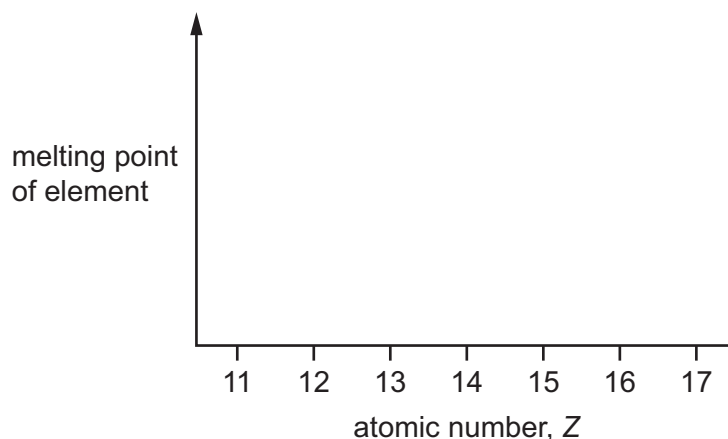
Use C = covalent, I = ionic, M = metallic, as appropriate.

[1]

(iv) Complete the row in the table labelled 'maximum oxidation number'.

[1]

- (b) Use the axes to sketch a graph that shows the trend in melting points of the elements with atomic numbers 11 to 17.



[2]

- (c) Dmitri Mendeleev published the first Periodic Table in 1869.

Mendeleev used his knowledge of chemical periodicity to propose the properties of gallium, $_{31}\text{Ga}$, a Group 13 element.

Table 2 gives some chemical and physical data of elements in Group 13.

Table 2

element	density / g cm^{-3}	boiling point /K	cationic radius /nm
$_5\text{B}$	2.34	3930	0.020
$_{13}\text{Al}$		2470	0.050
$_{31}\text{Ga}$	5.91	2400	
$_{49}\text{In}$	7.30		0.081
$_{81}\text{Tl}$	11.8	1460	0.095

Complete the table by predicting values for the missing data.

[3]

- (d) Indium and aluminium are elements in Group 13 of the Periodic Table.

Indium has very similar chemical properties to aluminium.

- Indium reacts vigorously with hydrochloric acid to form a colourless gas and a salt in solution.
- Indium oxide, In_2O_3 , is amphoteric.
- Gaseous indium bromide has the formula In_2Br_6 . This molecule contains coordinate bonds.

- (i) Identify the formula of the salt formed when indium reacts with hydrochloric acid.

[1]

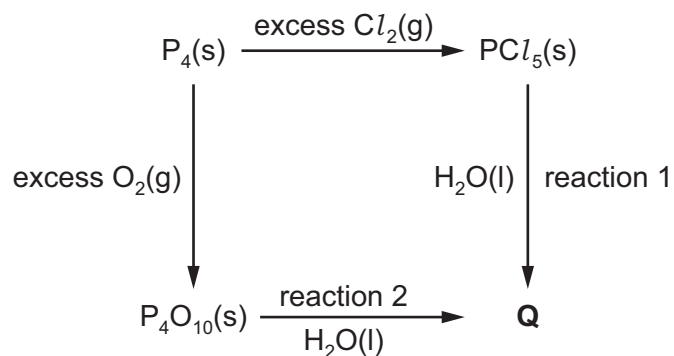
- (ii) Construct an equation for the reaction of In_2O_3 with excess aqueous NaOH .

[1]

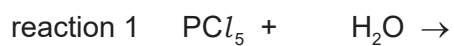
43. O/N 21/P23/Q3/b(ii)

3 Phosphorus is a reactive Period 3 element.

(b) Some reactions of $P_4(s)$ are shown in the reaction scheme.



(ii) Deduce the identity of **Q** and hence construct chemical equations for reactions 1 and 2.



[2]

44. M/J 21/P23/Q3/a

- 3** Separate samples of **R**, **S**, **T** and **U** are added to cold water. The identity of each sample is unknown. However, each sample is known to be pure and can only be one of Ba(OH)_2 , NaCl , P_4O_{10} or SiCl_4 .

- (a) (i)** Use the observations in the table to identify each sample as one of Ba(OH)_2 , NaCl , P_4O_{10} and SiCl_4 . Write your answers in the table.

	state at room temperature	observations on addition of sample to water	identity of sample
R	solid	alkaline, colourless solution is produced, some white solid remains	
S	solid	white solid disappears, solution is neutral	
T	liquid	misty fumes produced, white solid is made in vigorous reaction	
U	solid	acidic, colourless solution produced in vigorous reaction	

[4]

- (ii)** Identify the formula of the white solid made when sample **T** reacts with water.

[1]

- (iii)** Name the solution formed when sample **U** reacts with water.

[1]

45. MAR 21/P22/Q2/e(ii)

- 2** Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

- (e)** Aluminium reacts with chlorine to form aluminium chloride.

Aluminium chloride can exist as the gaseous molecule $\text{Al}_2\text{Cl}_6(\text{g})$. This molecule contains coordinate bonds.

- (ii)** Describe what you would see when solid aluminium chloride reacts with water.

Name the type of reaction that occurs.

[2]

46. O/N 20/P21/Q2/a(i,ii),b(ii),e

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(a) Phosphorus reacts with an excess of oxygen to form phosphorus(V) oxide.

(i) Write an equation to show the reaction of phosphorus with excess oxygen.

[1]

(ii) Describe the reaction of phosphorus(V) oxide with water.

[2]

(b) The two most common oxides of sulfur are SO_2 and SO_3 .

When SO_2 dissolves in water, a small proportion of it reacts with water to form a weak Brønsted-Lowry acid.

(ii) Write the equation for the reaction of SO_2 with water.

[1]

(e) Element **E** is a Period 5 element.

E reacts with oxygen to form an insoluble white oxide that has a melting point of 1910°C . The oxide of **E** conducts electricity only when liquid.

E also reacts readily with $\text{Cl}_2(\text{g})$ to form a white solid that reacts exothermically with water. The resulting solution reacts with aqueous silver nitrate to form a white precipitate that dissolves in dilute ammonia.

(i) Suggest the type of bonding shown by the **oxide** of **E**. Explain your answer.

[2]

(ii) Suggest the type of bonding shown by the **chloride** of **E**. Explain your answer.

[2]

47. O/N 20/P21/Q3/b

3 The reducing agent LiAlH_4 can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

(b) LiAlH_4 cannot be used in aqueous solution because it reacts with water to produce $\text{LiOH}(\text{aq})$, $\text{H}_2(\text{g})$ and a white precipitate which is soluble in excess sodium hydroxide.

Identify the white precipitate.

[1]

48. O/N 20/P22/Q2/a,b,c,e,g

2 The Period 3 elements, Na to S, all react with oxygen to form oxides.

(a) State the trend in acid/base behaviour of the oxides of the Period 3 elements, from Na to S.

[1]

(b) State and explain the trend, from Na to S, in the maximum oxidation number of the Period 3 elements in their oxides.

[2]

(c) Sodium oxide and phosphorus(V) oxide both react with water.

Name the product of each reaction.

reaction	product
sodium oxide with water	
phosphorus(V) oxide with water	

[2]

(e) Aluminium oxide, Al_2O_3 , reacts separately with both acids and alkalis.

(i) Write an equation for the reaction of aluminium oxide with excess aqueous hydrochloric acid.

[1]

(ii) Write an equation for the reaction of aluminium oxide with excess aqueous sodium hydroxide.

[1]

(g) Sodium oxide and silicon(IV) oxide react to form sodium silicate(IV), Na_2SiO_3 .

Sodium oxide is obtained from the thermal decomposition of sodium carbonate.

Write equations for the following reactions:

(i) sodium oxide with silicon(IV) oxide

[1]

(ii) the thermal decomposition of sodium carbonate, forming sodium oxide and carbon dioxide.

[1]

49. O/N 20/P22/Q3/a(i),b

3 PCl_5 , PCl_3 and NCl_3 are halides of Group 15 elements.

(a) PCl_5 can be formed from the reaction of phosphorus with chlorine. PCl_5 has a melting point of 161°C .

(i) Write an equation for the formation of PCl_5 from the reaction of phosphorus and chlorine.

[1]

(b) A small amount of PCl_5 is added to excess water. The PCl_5 reacts vigorously to form a colourless solution.

(i) Give **one** other observation you would make when PCl_5 reacts with excess water.

[1]

(ii) Write the equation for the reaction of PCl_5 with excess water.

[1]

(iii) Estimate the pH of the resulting solution.

[1]

50. O/N 20/P23/Q2/a(i,ii),b(ii),e

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(a) Phosphorus reacts with an excess of oxygen to form phosphorus(V) oxide.

(i) Write an equation to show the reaction of phosphorus with excess oxygen.

[1]

(ii) Describe the reaction of phosphorus(V) oxide with water.

[2]

(b) The two most common oxides of sulfur are SO_2 and SO_3 .

When SO_2 dissolves in water, a small proportion of it reacts with water to form a weak Brønsted-Lowry acid.

(ii) Write the equation for the reaction of SO_2 with water.

[1]

(e) Element **E** is a Period 5 element.

E reacts with oxygen to form an insoluble white oxide that has a melting point of 1910°C . The oxide of **E** conducts electricity only when liquid.

E also reacts readily with $\text{Cl}_2(\text{g})$ to form a white solid that reacts exothermically with water. The resulting solution reacts with aqueous silver nitrate to form a white precipitate that dissolves in dilute ammonia.

(i) Suggest the type of bonding shown by the **oxide** of **E**. Explain your answer.

[2]

(ii) Suggest the type of bonding shown by the **chloride** of **E**. Explain your answer.

[2]

51. O/N 20/P23/Q3/b

3 The reducing agent LiAlH_4 can be synthesised by reacting aluminium chloride with lithium hydride, LiH .

(b) LiAlH_4 cannot be used in aqueous solution because it reacts with water to produce $\text{LiOH}(\text{aq})$, $\text{H}_2(\text{g})$ and a white precipitate which is soluble in excess sodium hydroxide.

Identify the white precipitate.

[1]

52. M/J 20/P21/Q1/b(i),c

1 Gallium is a metal in Group 13 of the Periodic Table.

(b) Gallium and its compounds show similar properties to aluminium and its compounds. Gallium reacts with excess chlorine to form gallium trichloride.

(i) At 500 °C, gallium trichloride is a gas.

Suggest the type of attraction that exists at 500 °C

- between atoms within a gallium trichloride molecule
- between gallium trichloride molecules.

[2]

(c) Gallium metal reacts rapidly when exposed to air. A white solid layer is formed on its surface.

(i) Suggest an equation to describe the reaction occurring when gallium metal is exposed to air.

[2]

(ii) The table gives the formula of each gallium-containing product formed when gallium oxide reacts separately with hot aqueous hydrochloric acid and hot aqueous sodium hydroxide.

	formula of gallium-containing product
hot aqueous hydrochloric acid	GaCl_3
hot aqueous sodium hydroxide	NaGa(OH)_4

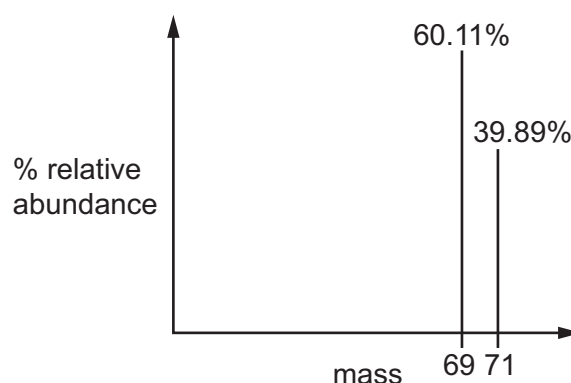
Give the name of the type of behaviour shown by gallium oxide in these reactions.

[1]

53. M/J 20/P22/Q1/e

- 1 Gallium is an element in Group 13.

A sample of gallium is analysed using a mass spectrometer. The mass spectrum produced is shown.



- (e) Gallium oxide, Ga_2O_3 , and aluminium oxide react in the same way with $\text{HCl}(\text{aq})$ and with $\text{NaOH}(\text{aq})$.

- (i) Suggest the equation for the reaction between Ga_2O_3 and $\text{HCl}(\text{aq})$.

[1]

- (ii) Suggest an equation for the reaction between gallium oxide and $\text{NaOH}(\text{aq})$.

[2]

54. MAR 20/P22/Q1/e

- 1 Group 2 metals form alkaline solutions in water.

- (e) Unlike the other oxides of Group 2 metals, beryllium oxide is amphoteric.

- (i) Give the meaning of the term *amphoteric*.

[1]

- (ii) Beryllium oxide and aluminium oxide have similar chemical properties.

The $\text{Be}(\text{OH})_4^{2-}$ anion is a product of the reaction between beryllium oxide and excess concentrated $\text{OH}^-(\text{aq})$.

Construct an equation for this reaction.

[1]

55. MAR 20/P22/Q2/b(ii)

- 2** The Group 17 elements, chlorine, bromine and iodine, are non-metals that show trends in their physical and chemical properties.

(b) The Group 17 elements can oxidise many metals to form halides.

- (ii)** Chlorine reacts with hot tin metal to form tin(IV) chloride, SnCl_4 .

SnCl_4 is a colourless liquid at room temperature that reacts vigorously with water to form an acidic solution.

Suggest the type of structure and bonding shown by SnCl_4 . Explain your answer.

[2]

56. O/N 19/P21/Q2/a,b(ii,iii),c(ii,iii)

- 2 (a) Complete the table to give details of the type of bonding and structure shown by some of the oxides of Period 3 elements.

	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	SO ₃
boiling point/°C	1275	3670	2977	2950	45
nature of oxide	basic	basic	amphoteric	acidic	acidic
bonding					
structure					

[2]

- (ii) Al₂O₃ is an amphoteric oxide.

Explain what is meant by the term *amphoteric*. Use chemical equations to illustrate your answer.

[3]

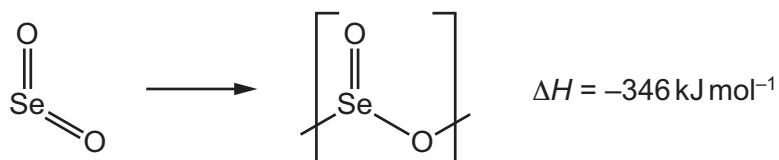
- (iii) State what you would observe when a small sample of Na₂O is placed in water.

[1]

- (c) Selenium is a Group 16 element which shows similar chemical reactions to sulfur.

- (ii) The most stable oxide of selenium is SeO₂.

Gaseous SeO₂ reacts to form a solid polymer, as shown. In the reaction one Se=O is replaced by two Se–O to form a polymer.



The bond enthalpy of Se=O is 514 kJ mol⁻¹.

Use these data to calculate the bond enthalpy, in kJ mol⁻¹, of Se–O.

bond enthalpy of Se–O = kJ mol⁻¹
[2]

- (iii) SeO₂ shows similar chemical reactions to SO₂.

Suggest an equation to show the reaction of SeO₂ with aqueous sodium hydroxide, NaOH.

[1]

57. O/N 19/P22/Q2/a,b(i)

- 2** Oxygen is the most abundant element in the Earth's crust. It reacts with other elements to form stable compounds, ions and molecules.

(a) Complete the table to give the formulae and acid/base behaviour of some of the oxides of the Period 3 elements.

element	sodium	aluminium	silicon	phosphorus	sulfur
formula of oxide	Na ₂ O				SO ₃
acid/base behaviour		amphoteric			

[2]

(b) Group 2 elements form stable hydroxides, with general formula M(OH)₂, where M is the Group 2 element.

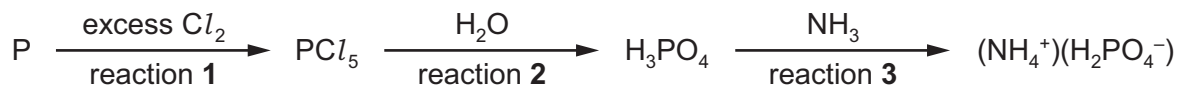
- (i)** Beryllium hydroxide, Be(OH)₂, is an amphoteric compound that shows similar chemical reactions to aluminium oxide.

State the meaning of the term *amphoteric*.

[1]

58. O/N 19/P22/Q3/a(i,ii)

- 3** A series of reactions for phosphorus and its compounds is shown.



- (a) (i)** State what you would observe in reaction 1.

[1]

- (ii)** State the type of reaction that occurs in reaction 2.

[1]

59. O/N 19/P23/Q2/a,b(ii,iii),c(ii,iii)

- 2 (a)** Complete the table to give details of the type of bonding and structure shown by some of the oxides of Period 3 elements.

	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	SO ₃
boiling point/°C	1275	3670	2977	2950	45
nature of oxide	basic	basic	amphoteric	acidic	acidic
bonding					
structure					

[2]

- (ii)** Al₂O₃ is an amphoteric oxide.

Explain what is meant by the term *amphoteric*. Use chemical equations to illustrate your answer.

[3]

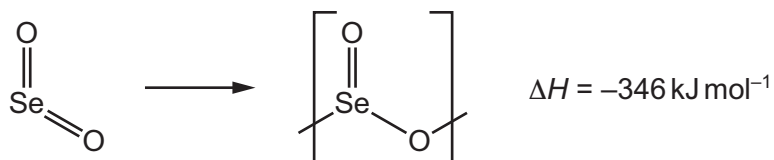
- (iii)** State what you would observe when a small sample of Na₂O is placed in water.

[1]

- (c)** Selenium is a Group 16 element which shows similar chemical reactions to sulfur.

- (ii)** The most stable oxide of selenium is SeO₂.

Gaseous SeO₂ reacts to form a solid polymer, as shown. In the reaction one Se=O is replaced by two Se–O to form a polymer.



The bond enthalpy of Se=O is 514 kJ mol⁻¹.

Use these data to calculate the bond enthalpy, in kJ mol⁻¹, of Se–O.

bond enthalpy of Se–O = kJ mol⁻¹
[2]

- (iii)** SeO₂ shows similar chemical reactions to SO₂.

Suggest an equation to show the reaction of SeO₂ with aqueous sodium hydroxide, NaOH.

[1]

60. M/J 19/P21/Q1/b

(b) Element **X** is a metal. **X** reacts with oxygen to form a black solid oxide. The oxidation state of **X** in this oxide is +2. The carbonate of **X**, XCO_3 , is a green solid. It decomposes on heating to form the oxide and a colourless gas.

(i) From the information given, state two similarities and one difference that metal **X** and its compounds have with Group 2 metals and their compounds.

similarity 1

similarity 2

difference 1

[3]

(ii) Write the formula of the oxide of **X**.

[1]

(iii) Write an equation for the reaction of XCO_3 when it is heated.

[1]

61. M/J 19/P21/Q2/b,f

2 Magnesium silicide, Mg_2Si , is a compound made by heating magnesium with sand.

(b) When solid Mg_2Si is added to water, silane gas, SiH_4 , and a solution of magnesium hydroxide are produced.

Construct the equation for this reaction. Include state symbols.

[2]

(f) Silicon dioxide reacts with hot, concentrated sodium hydroxide.

(i) Identify the **two** products formed during this reaction.

[2]

(ii) Describe the behaviour of the silicon dioxide during this reaction.

[1]

62. M/J 19/P22/Q3/a(i,ii,iii)

3 Period 3 elements react with chlorine gas, $\text{Cl}_2(\text{g})$, to form chlorides.

(a) The table shows the differences in observations which occur when two Period 3 chlorides are added to water.

Period 3 chloride	observations when added to water	pH of solution formed with water
NaCl	White solid disappears. Colourless solution made.	7
SiCl_4	Pale yellow solution forms. Bubbles form and the test-tube feels hot. White precipitate forms.	1–2

(i) Write an equation for the reaction occurring when SiCl_4 is added to cold water. Include state symbols.

[1]

(ii) Name the type of reaction occurring when SiCl_4 is added to water. Ignore the exothermic/endothermic nature of the reaction.

[1]

(iii) Explain, in terms of bonding, why NaCl and SiCl_4 behave differently when added to water.

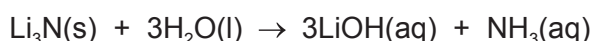
[2]

63. MAR 19/P22/Q1/b(ii)

1 Nitrogen, N_2 , is the most abundant gas in the Earth's atmosphere and is very unreactive.

(b) Magnesium and lithium both form nitrides with N_2 . These compounds both contain the N^{3-} ion.

(ii) Solid lithium nitride, Li_3N , reacts with water according to the following equation.



State **one** observation you would make during this reaction.

[1]

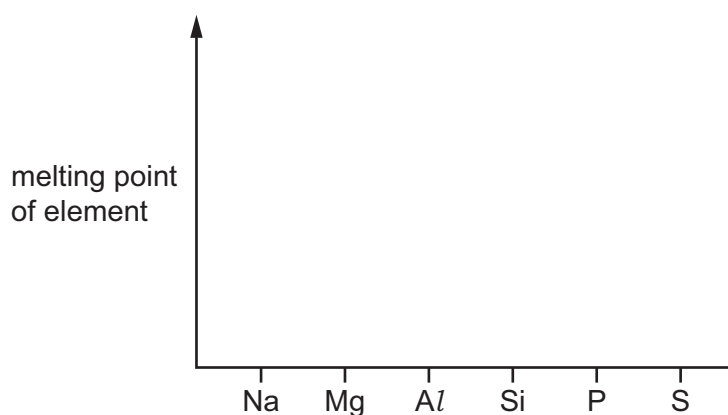
64. O/N 18/P22/Q2/a,c,d

2 The table gives some data for elements in the third period and some of their compounds.

element	Na	Mg	Al	Si	P	S
type of bonding	metallic				covalent	covalent
formula of oxide					P ₄ O ₁₀	SO ₂
formula of chloride	NaCl	MgCl ₂				SCl ₂

(a) Complete the table to show the bonding in the elements, and the formulae of their oxides and chlorides. [3]

(c) (i) On the axes, sketch the trend in melting point of the elements Na to S.



[1]

(ii) Give three statements to explain your sketch.

1

2

3

[3]

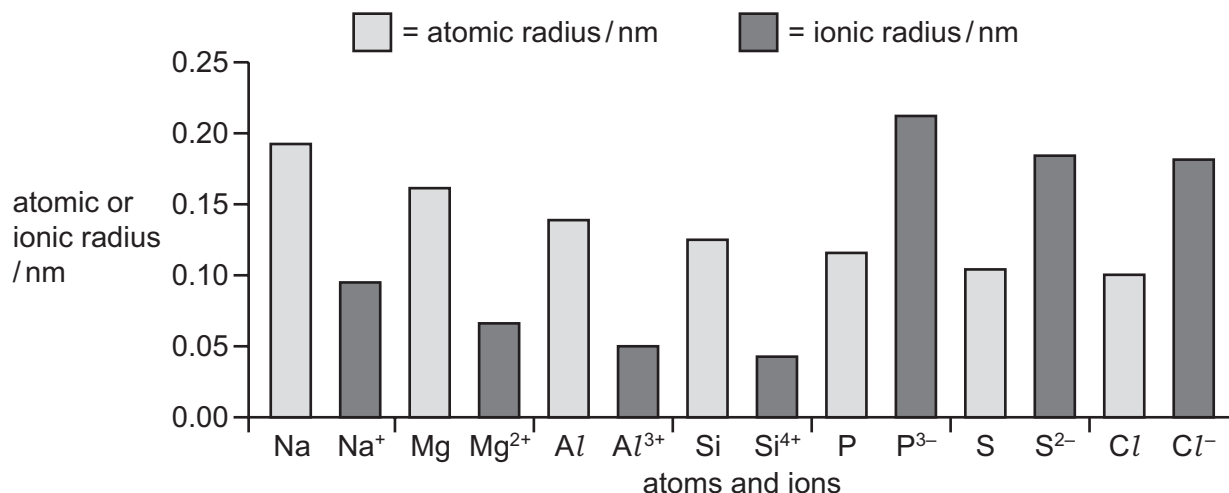
(d) Write an equation for the reaction of P₄O₁₀ with water.

[1]

65. M/J 18/P21/Q3/a(i),c

3 The elements in the third period exhibit periodicity in both their chemical and physical properties.

(a) A graph of the atomic and ionic radii across the third period is shown.



(i) Explain the decrease in atomic radius across the third period.

[2]

(c) The elements in the third period, from sodium to silicon, can react with chlorine to form chlorides.

(i) State and explain the pattern of change of oxidation number which occurs to both chlorine and the different Period 3 elements when they react together.

[3]

(ii) Give the equations to show the reactions of sodium chloride and silicon(IV) chloride when separately added to water.

sodium chloride

silicon(IV) chloride

[2]

(iii) Complete the table to describe the structure and bonding in sodium chloride and silicon(IV) chloride.

	structure	bonding
sodium chloride		
silicon(IV) chloride		

[2]

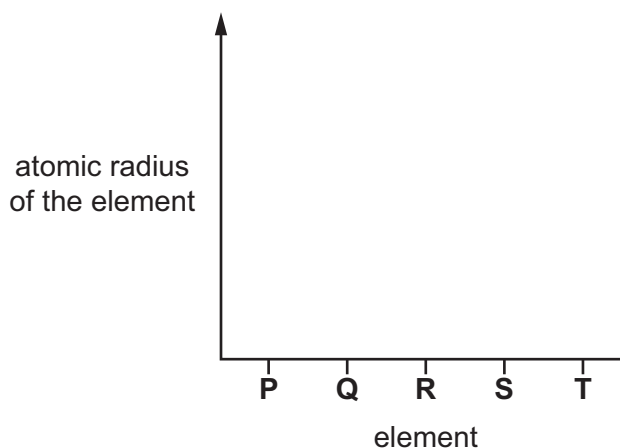
66. MAR 18/P22/Q1/b

(b) P–T are successive elements in Period 3 of the Periodic Table.

The letters are **not** the symbols of the elements.

On the axes, sketch a graph to show the trend in the atomic radius of the elements P–T.

Explain your answer.



explanation

[3]

67. MAR 18/P22/Q2/d(iii,v)

2 Carbon and silicon are elements in Group 14.

(d) Silicon shows the same kind of bonding and structure as diamond.

(iii) Solid Mg_2Si reacts with dilute hydrochloric acid to form gaseous SiH_4 and a solution of magnesium chloride.

Write an equation to show the reaction of solid Mg_2Si with dilute hydrochloric acid.

Include state symbols.

[2]

(v) SiH_4 reacts spontaneously with oxygen to produce a white solid and a colourless liquid that turns anhydrous copper(II) sulfate blue. No other products are formed.

Write an equation for the reaction of SiH_4 with oxygen.

State symbols are **not** required.

[1]

68. O/N 17/P21/Q2/b

- 2** The elements in the third period, and their compounds, show trends in their physical and chemical properties.
- (b)** The chlorides of the elements in the third period behave in different ways when added to water, depending on their structure and bonding.

L and **M** are each a chloride of an element in Period 3. A student investigated **L** and **M** and their results are given.

L is a white crystalline solid with a melting point of 987 K. **L** dissolves in water to form an approximately neutral solution. Addition of NaOH(aq) to an aqueous solution of **L** produces a white precipitate.

M is a liquid with a boiling point of 331 K. **M** is hydrolysed rapidly by cold water to form a strongly acidic solution, a white solid and white fumes.

Identify **L** and **M**.

Explain any properties and observations described.

Give equations where appropriate.

- (i)** **L** is

[3]

- (ii)** **M** is

[3]

69. O/N 17/P23/Q2/b

- 2** The elements in the third period, and their compounds, show trends in their physical and chemical properties.
- (b)** The chlorides of the elements in the third period behave in different ways when added to water, depending on their structure and bonding.

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M is a liquid with a boiling point of 331 K. **M** is hydrolysed rapidly by cold water to form a strongly acidic solution, a white solid and white fumes.

Identify **L** and **M**.

Explain any properties and observations described.

Give equations where appropriate.

- (i)** **L** is

[3]

- (ii)** **M** is

[3]

70. M/J 17/P21/Q3/a(i)

- 3 The properties of elements and their compounds show similarities, differences and trends depending on the positions of the elements in the Periodic Table.
- (a) The positions of some elements are indicated. The letters used are **not** the symbols of the elements.

[illegible]

From the elements labelled, give the letter for;

- (i) the element that forms an amphoteric oxide,

[1]

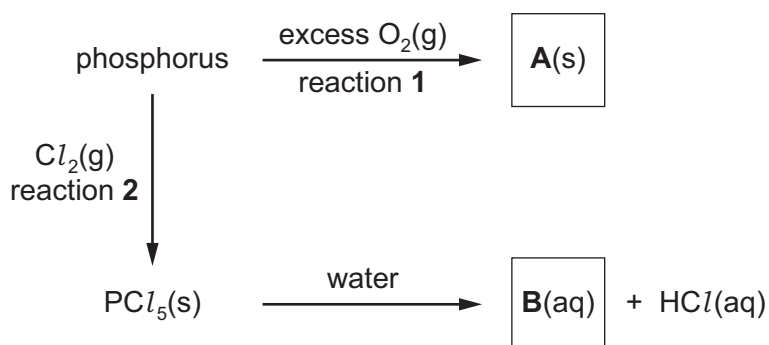
71. MAR 17/P22/Q1/a(i,ii),b

- 1 (a)** The table shows information about some of the elements in the third period.

element	Na	Mg	Al	P	S	Cl
atomic radius/nm	0.186	0.160	0.143	0.110	0.104	0.099
radius of most common ion/nm	0.095	0.065	0.050	0.212	0.184	0.181
maximum oxidation number of the element in its compounds	+1					+7

- (i) Complete the table to show the maximum oxidation number of each element in its compounds. [1]
- (ii) Explain why the atomic radius of elements in the third period decreases from Na to Cl. [3]
- (b)** Phosphorus is a non-metal in the third period. It reacts vigorously with excess oxygen but slowly with chlorine.

Some reactions of phosphorus are shown.

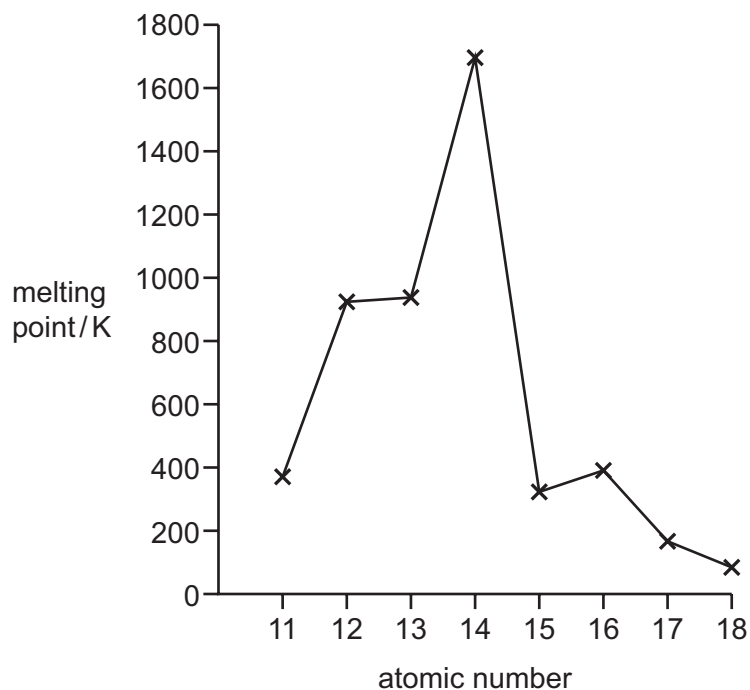


- (i) Write an equation to represent reaction 1, the formation of compound A. [1]
- (ii) Give **two** observations you could make in reaction 2. [2]
- 1.
 - 2.
- (iii) Name compound B. [1]

72. O/N 16/P22/Q3/b

3 The Periodic Table is arranged such that the properties of the elements show a number of trends.

(b) A plot of the melting points of the elements across the third period is shown.



- (i) Explain the increase in melting point from atomic number 11 to 12. [2]
- (ii) Suggest a reason why the increase from atomic number 12 to 13 is much smaller than the increase from atomic number 11 to 12. [1]
- (iii) State and explain the pattern of the melting points from atomic number 15 to 18. [3]
- (iv) Explain why the element with atomic number 14 has a melting point so much higher than the rest of the elements in the third period. [1]

73. M/J 16/P21/Q2/a,b,c,d

- 2** **D**, **E**, **F**, and **G** are four consecutive elements in the **fourth** period of the Periodic Table. (The letters are **not** the actual symbols of the elements.)

D is a soft, silvery metal with a melting point just above room temperature. Its amphoteric oxide, D_2O_3 , has a melting point of 1900°C and can be formed by heating **D** in oxygen.

G is a solid that can exist as several different allotropes, most of which contain G_8 molecules. **G** burns in air to form GO_2 which dissolves in water to form an acidic solution. This solution reacts with sodium hydroxide to form the salt Na_2GO_3 .

- (a) Suggest the identities of **D** and **G**.

D

G

[1]

- (b) Write equations for the reactions of D_2O_3 with

(i) hydrochloric acid,

[2]

(ii) sodium hydroxide.

[2]

- (c) Suggest the type of bonding and structure in D_2O_3 .

[1]

- (d) Write an equation for the formation of an acidic solution when GO_2 dissolves in water.

[1]

74. M/J 16/P22/Q2/d

- 2 The elements in Group 17, the halogens, and their compounds, show many similarities and trends in their properties. Some data are given for the elements fluorine to iodine.

element	bond energy /kJ mol ⁻¹	standard enthalpy change of atomisation, $\Delta H_{\text{at}}^{\ominus}$ /kJ mol ⁻¹	boiling point of element /K	boiling point of hydrogen halide /K
fluorine, F–F	158	79	85	293
chlorine, Cl–Cl	242	121	238	188
bromine, Br–Br	193	112	332	206
iodine, I–I	151	107	457	238

- (d) Chlorine reacts directly with many elements to form chlorides. Three such compounds are MgCl_2 , AlCl_3 and SiCl_4 .

- (i) State and explain the pattern shown by the formulae of these three chlorides.

[2]

- (ii) Write equations to show the behaviour of each of these chlorides when added to water.



[3]

75. M/J 16/P23/Q1/b(ii)

- 1 An experiment was carried out to determine the percentage of iron in a sample of iron wire.

- (b) Some electronegativity values are shown.

element	electronegativity
aluminium	1.5
chlorine	3.0
iron	1.8

- (ii) Suggest an equation for the reaction between iron(III) chloride and water.

[1]

76. MAR 16/P22/Q1/a(i,iii)

1 This question is about Period 3 elements and their compounds.

(a) Give an explanation for each of the following statements.

(i) The atomic radius decreases across Period 3 (Na to Ar).

[2]

(iii) Sodium is a better electrical conductor than phosphorus.

[2]

77. O/N 15/P22/Q3/a,b

3 The elements in Period 3, Na, Mg, Al, P and S, all react with oxygen when heated in air.

- (a) (i)** Give the formula of the oxide formed when each element is heated in air. One has been completed for you.

Na =

Mg =

Al = Al_2O_3

P =

S =

[2]

- (ii)** Describe what you would **see** when sodium and sulfur are each heated separately in air and give an equation for each reaction.

Na

equation

S

equation

[4]

(b) The oxides show variations in their behaviour when added to water, acids and alkalis.

- (i)** Place the symbols of the elements in **(a)(i)** in the appropriate row of the table to indicate this behaviour.

acidic	
amphoteric	
basic	

[2]

- (ii)** State the bonding present in acidic and basic oxides.

acidic

basic

[2]

- (iii)** Write equations for the reaction of aluminium oxide with each of hydrochloric acid, HCl, and sodium hydroxide, NaOH.

with HCl

with NaOH

[2]

78. M/J 15/P21/Q1/c(ii,iii),d

- (c) The element tellurium, Te, reacts with chlorine to form a single solid product, with a relative formula mass of 270. The product contains 52.6% chlorine by mass.

- (ii) This chloride melts at 224 °C and reacts vigorously with water.

State the type of bonding **and** structure present in this chloride and explain your reasoning.

[2]

- (iii) Suggest an equation for the reaction of this chloride with water.

[1]

- (d) Sodium and silicon also react directly with chlorine to produce the chlorides shown.

chloride	melting point/°C	difference between the electronegativities of the elements
NaCl	801	2.2
SiCl ₄	−69	1.3

- (i) Describe what you would **see** during the reaction between sodium and chlorine.

[2]

- (ii) Explain the differences between the melting points of these two chlorides in terms of their structure **and** bonding. You should refer to the difference between the electronegativities of the elements in your answer.

NaCl structure **and** bonding

SiCl₄ structure **and** bonding

explanation

[4]

79. M/J 15/P22/Q1/d

- (d) The acid/base behaviour of the oxides in the third period varies across the period.

- (i) Describe this behaviour and explain it with reference to the structure and bonding of sodium oxide, Na₂O, aluminium oxide, Al₂O₃, and sulfur trioxide, SO₃.

[2]

- (ii) Write equations for reactions of these three oxides with hydrochloric acid and/or sodium hydroxide as appropriate.

[4]

Answers Section

1. M/J 26/P22/Q1/c,d

1(c)(i)	sodium oxide	1	I formulas BOD sodium
1(c)(ii)	$\text{Li}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{LiOH}$	1	A $2\text{Li}^+ + 2\text{OH}^-$ for 2LiOH BOD LiHO I equations with water as both a reactant and a product
1(c)(iii)	M1 phosphorus (V) chloride / PCl_5 M2 aluminium chloride / AlCl_3	2	M1 A phosphorus pentachloride M1 I phosphorus chloride / phosphorus trichloride M1 I phosphorus alone M2 A Al_2Cl_6 M2 A aluminium alone / aluminium trichloride / aluminium(III) chloride M2 I incorrect aluminium chloride formula I MgCl_2 / magnesium chloride / magnesium I sulfur chlorides / sulfur R silicon chlorides / silicon R NaCl / Na R any period 3 chloride with additional elements R any chloride not from period 3
1(d)	nitrogen dioxide OR NO_2	1	A nitrogen(IV) oxide I nitrogen oxide alone

2. M/J 26/P23/Q1/a

1(a)(i)	<table border="1"> <thead> <tr> <th>element</th><th>formula of chloride</th><th>type of bonding</th><th>type of reaction, if any of the chloride with cold water</th></tr> </thead> <tbody> <tr> <td>sodium</td><td>NaCl</td><td>ionic</td><td>no reaction / none A dissolves / dissociates / dissolution</td></tr> <tr> <td>silicon</td><td>SiCl_4</td><td>covalent</td><td>hydrolysis R hydration</td></tr> </tbody> </table>	element	formula of chloride	type of bonding	type of reaction, if any of the chloride with cold water	sodium	NaCl	ionic	no reaction / none A dissolves / dissociates / dissolution	silicon	SiCl_4	covalent	hydrolysis R hydration	3	•✓•✓•✓ I giant / molecular in 'bonding'
element	formula of chloride	type of bonding	type of reaction, if any of the chloride with cold water												
sodium	NaCl	ionic	no reaction / none A dissolves / dissociates / dissolution												
silicon	SiCl_4	covalent	hydrolysis R hydration												
1(a)(ii)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1	I $\text{PCl}_5 + \text{H}_2\text{O} \rightarrow \text{POCl}_3 + 2\text{HCl}$												

3. M/J 26/P24/Q3/a,b

3(a)(i)	decreases (across the period)	1	A smaller / goes down A decreases down the period R decreases down the group																		
3(a)(ii)	Na ₂ O	1	A Na ₂ O ₂ and NaO ₂ A equation that gives Na ₂ O as the only product I balancing																		
3(a)(iii)	MgO + H ₂ O → Mg(OH) ₂	1	A MgO + H ₂ O → Mg ²⁺ + 2OH ⁻ A in equation =																		
3(b)(i)	<table><tr><td></td><td>type of bonding</td><td>type of structure</td></tr><tr><td>NaCl</td><td>I (ionic)</td><td>G (giant)</td></tr><tr><td>SiCl₄</td><td>C (covalent)</td><td>S (simple)</td></tr></table>		type of bonding	type of structure	NaCl	I (ionic)	G (giant)	SiCl ₄	C (covalent)	S (simple)	2	●✓●✓ A <table><tr><td></td><td>type of bonding</td><td>type of structure</td></tr><tr><td>NaCl</td><td>GI</td><td>GI</td></tr><tr><td>SiCl₄</td><td>SC</td><td>SC</td></tr></table>		type of bonding	type of structure	NaCl	GI	GI	SiCl ₄	SC	SC
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NaCl	I (ionic)	G (giant)																			
SiCl ₄	C (covalent)	S (simple)																			
	type of bonding	type of structure																			
NaCl	GI	GI																			
SiCl ₄	SC	SC																			

3(b)(ii)	M1 (formula:) H_3PO_4 M2 (pH of solution:) 0–4	2 M1 $\text{A PO}(\text{OH})_3$ M1 R H_3PO_3 (as excess water) M1 R HCl on formula answer line M1 R $\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$ on formula answer line
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4. O/N 25/P21/Q2/a,b,c

2(a)(i)	M1 nuclear charge increases (across period) AND similar shielding M2 greater / stronger attraction to nucleus	2														
2(a)(ii)	M1 Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3− M2 Al ⁽³⁺⁾ has (one) fewer shell (than P ^(3−))	2														
2(b)(i)	<table><tr><td>M1</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td><td>(+)6</td></tr><tr><td>M2</td><td>12-14</td><td>8-10</td><td>—</td><td>—</td><td>0-4</td><td>0-4</td></tr></table>	M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6	M2	12-14	8-10	—	—	0-4	0-4	2
M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6										
M2	12-14	8-10	—	—	0-4	0-4										
2(b)(ii)	they are insoluble	1														
2(c)(i)	Na ₂ O + 2HCl → 2NaCl + H ₂ O	1														
2(c)(ii)	Al ₂ O ₃ + 2NaOH → 2NaAlO ₂ + H ₂ O	1														

5. O/N 25/P22/Q2/a,b,c,d

2(a)(i)	delocalised electrons move (through the structure / between layers)					1	
2(a)(ii)	P / S / Cl have no charge carriers OR (no ions and) no delocalised electrons OR no mobile / moving ions					1	
2(b)	M1 Si has (lots of) strong covalent bonds broken M2 P / S / Cl (only) has weak Van der Waals' (forces) broken					2	
2(c)	M1	+1	+2	+3	+4	+5	4
	M2	7	6.5	0–4	0–4	0–4	
	M3	I	I	I	C	C	
	M4	G	G	G	S	S	
2(d)(i)	$2Al + 3Cl_2 \rightarrow 2AlCl_3$					1	
2(d)(ii)	$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$					1	

6. O/N 25/P23/Q2/a,b,c

2(a)(i)	M1 nuclear charge increases (across period) AND similar shielding M2 greater / stronger attraction to nucleus	2														
2(a)(ii)	M1 Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3− M2 Al (3+) has (one) fewer shell (than P(3−))	2														
2(b)(i)	<table><tr><td>M1</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td><td>(+)6</td></tr><tr><td>M2</td><td>12-14</td><td>8-10</td><td>—</td><td>—</td><td>0-4</td><td>0-4</td></tr></table>	M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6	M2	12-14	8-10	—	—	0-4	0-4	2
M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6										
M2	12-14	8-10	—	—	0-4	0-4										
2(b)(ii)	they are insoluble	1														

2(c)(i)	$\text{Na}_2\text{O} + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O}$	1
2(c)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$	1

7. O/N 25/P24/Q1/a(ii),d

1(a)(ii)	$2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$	1
1(d)(i)	M1 MgCl_2 (is ionic and) has mobile ions M2 SiCl_4 is covalent AND no charge carriers OR (no ions and) no delocalised electrons OR no mobile / moving ions	2
1(d)(ii)	M1 MgCl_2 pH = 7 M2 SiCl_4 pH 0 – 4	2

8. O/N 25/P24/Q2/c(iii)

2(c)(iii)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1
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9. M/J 25/P21/Q1/b,c

1(b)(i)	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$	1									
1(b)(ii)	$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	1									
1(b)(iii)	<table border="1"> <tr> <td></td><td>sodium</td><td>sodium oxide</td></tr> <tr> <td>one similarity</td><td colspan="2">(both) solids disappear OR (both produce) colourless solutions</td></tr> <tr> <td>one difference</td><td>effervescence</td><td>no effervescence</td></tr> </table>		sodium	sodium oxide	one similarity	(both) solids disappear OR (both produce) colourless solutions		one difference	effervescence	no effervescence	2
	sodium	sodium oxide									
one similarity	(both) solids disappear OR (both produce) colourless solutions										
one difference	effervescence	no effervescence									
1(c)	M1 $\text{NaCl} + 1$ AND $\text{PCl}_5 + 5$ M2 Na has only 1 valence (shell) electrons AND P has 5	2									

10. M/J 25/P22/Q1/b,c

1(b)(i)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$ $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1 1
1(b)(ii)	M1 phosphorus(V) chloride white / yellow solid M2 silicon(IV) chloride colourless liquid	2
1(b)(iii)	M1 (phosphorus(V) chloride) produces colourless solution M2 silicon(IV) chloride produces white suspension / white precipitate (in a colorless liquid / mixture / solution)	2
1(c)(i)	species behaves as acid and base OR species reacts with (both) acid and base	1
1(c)(ii)	Al_2O_3	1

11. M/J 25/P23/Q2/a,b,c(i),e

2(a)	Na_2O	1
2(b)	$\text{P}_4\text{O}_{10} + 12\text{NaOH} \rightarrow 4\text{Na}_3\text{PO}_4 + 6\text{H}_2\text{O}$	1
2(c)(i)	SO_3	1
2(e)(i)	$\text{NaAl}(\text{OH})_4$	1

2(e)(ii)	$\text{Al}_2(\text{SO}_4)_3$	1
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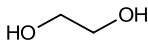
12. M/J 25/P24/Q1/b,d

1(b)	M1 Na to Al have (delocalised) electrons that can move (through the structure / lattice) AND (P / S / Cl) do not (have delocalised electrons / mobile charge carriers) M2 Na to Al increasing number of delocalised electrons	2
1(d)(i)	$\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
1(d)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$	1

13. M/J 25/P24/Q2/b(i,iii)

2(b)(i)	<i>In any order:</i> M1 (solution / liquid) (red) to colourless M2 yellow solid appears	2
2(b)(iii)	hydrolysis	1

14. MAR 25/P22/Q1/b(i,iii,iv,v)

1(b)(i)	green colour (of gas) disappears	1
1(b)(iii)	$\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
1(b)(iv)	phosphoric(V) acid	1
1(b)(v)		1

15. O/N 24/P21/Q3/e(i,ii),f

3(e)(i)	$\text{Si} + 2\text{Cl}_2 \rightarrow \text{SiCl}_4$	1
3(e)(ii)	effervescence / misty / steamy fumes	1
3(f)	$\text{Sn}(\text{SO}_4)_2$	1

16. O/N 24/P22/Q2/a(i,iii),b(i)

2(a)(i)	$\text{M1 } 4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$	1									
	$\text{M2 } \text{S} + \text{O}_2 \rightarrow \text{SO}_2$	1									
2(a)(iii)	it increases	1									
	as number of valence electrons (which can be used in bonding / lost / shared / donated) increases	1									
2(b)(i)	<table border="1"> <thead> <tr> <th>Period 3 oxide</th><th>product of rxn with water</th><th>pH of solution formed</th></tr> </thead> <tbody> <tr> <td>• MgO</td><td>$\text{Mg}(\text{OH})_2$</td><td>• $8 \leq \text{pH} \leq 12$</td></tr> <tr> <td>P_4O_{10}</td><td>• H_3PO_4</td><td>• $1 \leq \text{pH} \leq 4$</td></tr> </tbody> </table>	Period 3 oxide	product of rxn with water	pH of solution formed	• MgO	$\text{Mg}(\text{OH})_2$	• $8 \leq \text{pH} \leq 12$	P_4O_{10}	• H_3PO_4	• $1 \leq \text{pH} \leq 4$	2
Period 3 oxide	product of rxn with water	pH of solution formed									
• MgO	$\text{Mg}(\text{OH})_2$	• $8 \leq \text{pH} \leq 12$									
P_4O_{10}	• H_3PO_4	• $1 \leq \text{pH} \leq 4$									

•✓•✓

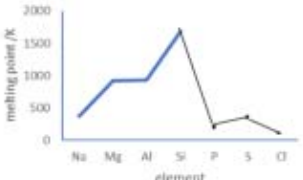
17. O/N 24/P23/Q3/e(i,ii),f

3(e)(i)	$\text{Si} + 2\text{Cl}_2 \rightarrow \text{SiCl}_4$	1
3(e)(ii)	effervescence / misty / steamy fumes	1
3(f)	$\text{Sn}(\text{SO}_4)_2$	1

18. M/J 24/P21/Q2/a

2(a)(i)	hydrolysis	1
2(a)(ii)	silicon(IV) chloride / silicon tetrachloride ALLOW phosphorus(III) chloride / phosphorus trichloride	1

19. M/J 24/P22/Q1/d,e,f

1(d)(i)	large amount of energy required to break bonds in giant covalent structure	1						
1(d)(ii)	 M1 all 3 melting points lower than Mg M2 show melting point Cl < P < S	2						
1(e)(i)	<table border="1" data-bbox="236 795 933 891"> <tr> <td>Na</td><td>solid</td><td>10–14</td></tr> <tr> <td>S</td><td>gas</td><td>0–4</td></tr> </table> 4 correct = 2 marks 2 or 3 correct = 1 mark 0 or 1 correct = 0 marks	Na	solid	10–14	S	gas	0–4	2
Na	solid	10–14						
S	gas	0–4						
1(e)(ii)	acid–base OR neutralisation	1						
1(e)(iii)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1						
1(f)(i)	reacts with both acids and bases OR shows both acidic and basic behaviour	1						
1(f)(ii)	$\text{Al}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaAl}(\text{OH})_4$	1						

20. M/J 24/P23/Q1/b,c

1(b)(i)	(+)III	1
1(b)(ii)	number of outer / valence electrons	1
1(c)(i)	both states shown as solid	1
1(c)(ii)	M1 correctly balanced equation for aluminium oxide $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ M2 correctly balanced equation for P_4O_{10} and an excess of NaOH $\text{P}_4\text{O}_{10} + 12\text{NaOH} \rightarrow 4\text{Na}_3\text{PO}_4 + 6\text{H}_2\text{O}$	2

21. MAR 24/P22/Q2/a,c(ii)

2(a)	formula of chloride	• NaCl	• MgCl ₂	• SiCl ₄	3
	structure	giant	• giant	• simple	
	bonding of chloride	• ionic	• ionic	covalent	
	pH of solution formed	• 7	6.2	• 1–4	

2(c)(ii)	H ₂ O / water	1
	hydrolysis	1

22. O/N 23/P21/Q1/d(ii)

1(d)(ii)	$\text{POCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 3\text{HCl}$	1
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23. O/N 23/P22/Q1/a

1(a)(i)	decreases (from Si to S)	1
1(a)(ii)	M1 Si giant covalent AND S is simple molecular OR Si giant covalent AND S has IMFs	1
	M2 Si involves breaking covalent bonds AND S involves breaking IMFs	1
	M3 more energy needed to break covalent bonds than IMFs OR covalent bonds are strong AND IMFs are weak	1

24. O/N 23/P22/Q3/a(i)

3(a)(i)	$4\text{P} / \text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1

25. O/N 23/P23/Q1/d(ii)

1(d)(ii)	$\text{POCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 3\text{HCl}$	1
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26. M/J 23/P21/Q1/f

1(f)(i)	$\text{TeF}_x + x\text{H}_2\text{O} \rightarrow \text{Te}(\text{OH})_x + x\text{HF}$ OR $\text{TeF}_6 + 6\text{H}_2\text{O} \rightarrow \text{Te}(\text{OH})_6 + 6\text{HF}$	1
1f(ii)	hydrolysis	1

27. M/J 23/P21/Q3/a,b

3(a)	M1 <i>appearance</i> (colourless / yellow) liquid M2 <i>structure and bonding</i> simple / molecular AND covalent			2																	
3(b)	<table><tr><td></td><td>MgCl₂</td><td>PCl₅</td><td rowspan="2">[1]</td></tr><tr><td>appearance</td><td>● (white) crystals / solid</td><td>● (white OR pale yellow) crystals / solid</td></tr><tr><td>one similarity in observation</td><td colspan="2">(colourless) solution made</td><td>[1]</td></tr><tr><td>one difference in observation</td><td>no (misty / steamy) fumes</td><td>misty / steamy fumes</td><td>[1]</td></tr><tr><td>pH</td><td>● 6.5–7</td><td>● 0–4</td><td>[1]</td></tr></table>		MgCl ₂	PCl ₅	[1]	appearance	● (white) crystals / solid	● (white OR pale yellow) crystals / solid	one similarity in observation	(colourless) solution made		[1]	one difference in observation	no (misty / steamy) fumes	misty / steamy fumes	[1]	pH	● 6.5–7	● 0–4	[1]	4
	MgCl ₂	PCl ₅	[1]																		
appearance	● (white) crystals / solid	● (white OR pale yellow) crystals / solid																			
one similarity in observation	(colourless) solution made		[1]																		
one difference in observation	no (misty / steamy) fumes	misty / steamy fumes	[1]																		
pH	● 6.5–7	● 0–4	[1]																		

28. M/J 23/P22/Q1/a(iii)

1(a)(iii)	M1 S_8 / molecules of sulfur have more electrons (than P_4 / molecules of phosphorus) M2 S has stronger instantaneous dipole–induced dipole forces (than phosphorus / P)	2
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29. M/J 23/P22/Q2/a,b,c

2(a)(i)		<table><tr><td>element</td><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>maximum oxidation number of elements in chlorides</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td></tr></table>	element	Na	Mg	Al	Si	P	maximum oxidation number of elements in chlorides	(+)1	(+)2	(+)3	(+)4	(+)5	1
	element	Na	Mg	Al	Si	P									
	maximum oxidation number of elements in chlorides	(+)1	(+)2	(+)3	(+)4	(+)5									
All correct for one mark															
2(a)(ii)	number of outer/valence electrons			1											

2(b)(i)	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
2(b)(ii)	1–4	1
2(c)(i)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow 5\text{HCl} + \text{H}_3\text{PO}_4$	1
2(c)(ii)	1–4	1

30. M/J 23/P23/Q2/a,c

2(a)(i)	it / aluminium oxide is insoluble (in water) OR it / aluminium oxide does not react (with water)	1
2(a)(ii)	M1 correct state of correct aluminium product M2 balanced equation for reaction of aluminium oxide with NaOH $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4(\text{aq})$ OR $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2(\text{aq}) + \text{H}_2\text{O}$	2
2(a)(iii)	sodium (only)	1
2(c)(i)	Al (+)3 / III AND P (+)5 / V	1
2(c)(ii)	different number of valence/outer shell electrons	1
2(c)(iii)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1
2(c)(iv)	hydrolysis / substitution	1
2(c)(v)	HCl(g) dissolves in water to make (hydrochloric) acid	1

31. MAR 23/P22/Q1/d(i,ii)

1(d)(i)	$\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$	1
1(d)(ii)	$\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$	1

32. O/N 22/P21/Q3/a

3(a)(i)	NaCl AND MgCl ₂	1
3(a)(ii)	AlCl ₃ AND SiCl ₄ AND PCl ₅	1
3(a)(iii)	NaCl	1
3(a)(iv)	SiCl ₄	1

33. O/N 22/P22/Q2/a,b

2(a)	<table><tr><td>NaCl</td><td>AlCl₃</td><td>SiCl₄</td><td>PCl₅</td></tr><tr><td>I</td><td>I OR C</td><td>C</td><td>C</td></tr></table>	NaCl	AlCl ₃	SiCl ₄	PCl ₅	I	I OR C	C	C	1
	NaCl	AlCl ₃	SiCl ₄	PCl ₅						
	I	I OR C	C	C						
	<table><tr><td>G</td><td>G</td><td>S</td><td>S</td></tr></table>	G	G	S	S	1				
G	G	S	S							
<table><tr><td>+1</td><td>+3</td><td>+4</td><td>+5</td></tr></table>	+1	+3	+4	+5	1					
+1	+3	+4	+5							
	<table><tr><td>+3</td><td>+2</td></tr></table>	+3	+2	1						
+3	+2									

2(b)	M1 NaCl(s) → Na ⁺ (aq) + Cl ⁻ (aq)	1
	PCl ₅ (s) + 4H ₂ O(l) → H ₃ PO ₄ (aq) + 5HCl(aq) M2 balanced equation	1
	M3 all state symbols for PCl ₅ equation	1

34. O/N 22/P23/Q3/a

3(a)(i)	NaCl AND MgCl_2	1
3(a)(ii)	AlCl_3 AND SiCl_4 AND PCl_5	1
3(a)(iii)	NaCl	1
3(a)(iv)	SiCl_4	1

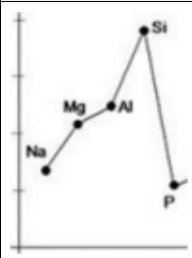
35. M/J 22/P21/Q1/f(i)

1(f)(i)	white flame / light OR white solid / smoke	1
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36. M/J 22/P21/Q3/d(iii)

3(d)(iii)	M1 water reacts with / hydrolyses PCl_5 M2 $\text{H}_2\text{O} + \text{PCl}_5 \rightarrow \text{POCl}_3 + 2\text{HCl}$ OR $4\text{H}_2\text{O} + \text{PCl}_5 \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	2
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37. M/J 22/P23/Q2/a

2(a)(i)				3												
2(a)(ii)	<table><tr><td></td><td>sodium chloride</td><td>phosphorus (V) chloride</td></tr><tr><td>state at room temperature</td><td>solid</td><td>solid</td></tr><tr><td>name of change which occurs on addition of water</td><td>dissolves</td><td>hydrolysis</td></tr><tr><td>pH of final solution</td><td>7</td><td>1–2</td></tr></table>				sodium chloride	phosphorus (V) chloride	state at room temperature	solid	solid	name of change which occurs on addition of water	dissolves	hydrolysis	pH of final solution	7	1–2	3
	sodium chloride	phosphorus (V) chloride														
state at room temperature	solid	solid														
name of change which occurs on addition of water	dissolves	hydrolysis														
pH of final solution	7	1–2														

38. MAR 22/P22/Q2/a(ii),b(i,ii),c(iv)

2(a)(ii)	$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	1
2(b)(i)	reacts with both acids and bases / shows both acidic and basic behaviour	1
2(b)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$	1
2(c)(iv)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1

39. MAR 22/P22/Q3/b(i)

3(b)(i)	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
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40. O/N 21/P21/Q3/b(ii)

3(b)(ii)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$ $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	2
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41. O/N 21/P22/Q1/f(ii)

1(f)(ii)	hydrolysis	1
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42. O/N 21/P22/Q2/a(i,iv),b,c,d(i,ii)

2(a)(i)	M M M C C C C	1
2(a)(iv)	(+1) (+2) (+3) (+4) (+5) (+6)	1
2(b)		2
2(c)	M1 density of ${}_{13}\text{Al}$: value within range 2.5–5.0 (g cm^{-3}) M2 cationic radius of ${}_{31}\text{Ga}$: value within range 0.055–0.075 (nm) M3 boiling point of ${}_{49}\text{In}$: value within range 1500–2300 (K)	3
2(d)(i)	InCl_3	1
2(d)(ii)	$\text{In}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaIn}(\text{OH})_4$	1

43. O/N 21/P23/Q3/b(ii)

3(b)(ii)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$ $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	2
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44. M/J 21/P23/Q3/a

3(a)(i)		state at room temp	observations on add'n of sample to water	identity of sample	4
	R	solid	alkaline, colourless solution is made but some white solid remains	M1 Ba(OH) ₂ OR barium hydroxide	
	S	solid	white solid disappears, solution is neutral	M2 NaCl OR sodium chloride	
	T	liquid	misty fumes, white solid is made in vigorous reaction	M3 SiCl ₄ OR silicon(IV) chloride	
	U	solid	acidic, colourless solution is made in vigorous reaction	M4 P ₄ O ₁₀ OR phosphorus(V) oxide	
3(a)(ii)	SiO ₂				1
3(a)(iii)	phosphoric acid / phosphoric(V)acid				1

45. MAR 21/P22/Q2/e(ii)

2(e)(ii)	M1: (AlCl_3 / solid) disappears - misty / steamy fumes - temperature increases M2: hydrolysis	2
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46. O/N 20/P21/Q2/a(i,ii),b(ii),e

2(a)(i)	$\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
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2(a)(ii)	any two from: - reacts vigorously - solid disappears / colourless solution forms - hydrolysis - exothermic - acid(ic) (solution) - steamy / misty fumes	2
2(b)(ii)	$\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$	1
2(e)(i)	M1: ionic M2: ions only able / free to move / free to conduct (when liquid / molten)	2
2(e)(ii)	M1: covalent M2: hydrolysed (by water)	2

47. O/N 20/P21/Q3/b

3(b)	$\text{Al}(\text{OH})_3$ / aluminium hydroxide	1
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48. O/N 20/P22/Q2/a,b,c,e,g

2(a)	more acidic / less basic (from Na to S across period)		1
2(b)	M1: increases (from Na to S / across period)		2
	M2: increasing (number of) valence electrons OR (number of) electrons in outer (electron) shell increases		
2(c)	reaction	name of product	2
	sodium oxide with water	sodium hydroxide	
	phosphorus(V) oxide with water	phosphoric(V) acid	
2(e)(i)	$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$		1
2(e)(ii)	$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$		1
2(g)(i)	$Na_2O + SiO_2 \rightarrow Na_2SiO_3$		1
2(g)(ii)	$Na_2CO_3 \rightarrow Na_2O + CO_2$		1

49. O/N 20/P22/Q3/a(i),b

3(a)(i)	$\text{P}_4 + 10\text{Cl}_2 \rightarrow 4\text{PCl}_5$	1
3(b)(i)	steamy / misty fumes	1
3(b)(ii)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1
3(b)(iii)	0 to 4	1

50. O/N 20/P23/Q2/a(i,ii),b(ii),e

2(a)(i)	$\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
2(a)(ii)	any two from: - reacts vigorously - solid disappears / colourless solution forms - hydrolysis - exothermic - acid(ic) (solution) - steamy / misty fumes	2
2(b)(ii)	$\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$	1

2(e)(i)	M1: ionic	2
	M2: ions only able / free to move / free to conduct (when liquid / molten)	
2(e)(ii)	M1: covalent	2
	M2: hydrolysed (by water)	

51. O/N 20/P23/Q3/b

3(b)	$\text{Al}(\text{OH})_3$ / aluminium hydroxide	1
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52. M/J 20/P21/Q1/b(i),c

1(b)(i)	M1 attractions between atoms within a gallium trichloride molecule covalent (bonds) M2 attractions between gallium trichloride molecules temporary induced dipoles	2
1(c)(i)	$4\text{Ga} + 3\text{O}_2 \rightarrow 2\text{Ga}_2\text{O}_3$ M1 correct formula of Ga_2O_3 M2 correctly balanced equation based on $\text{Ga} + \text{O}_2$ and formula of gallium oxide in M1	2
1(c)(ii)	amphoteric	1

53. M/J 20/P22/Q1/e

1(e)(i)	$\text{Ga}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{GaCl}_3 + 3\text{H}_2\text{O}$	1
1(e)(ii)	M1 Identity of correct gallium containing product $\text{NaGa}(\text{OH})_4$ OR NaGaO_2 M2 correctly balanced equation for reaction of Ga_2O_3 with $\text{NaOH}(\text{aq})$ EITHER $\text{Ga}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaGa}(\text{OH})_4$ OR $\text{Ga}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaGaO}_2 + \text{H}_2\text{O}$	2

54. MAR 20/P22/Q1/e

1(e)(i)	reacts with / behaves as both acid and base	1
1(e)(ii)	$\text{BeO} + 2\text{OH}^- + \text{H}_2\text{O} \rightarrow \text{Be}(\text{OH})_4^{2-}$	1

55. MAR 20/P22/Q2/b(ii)

2(b)(ii)	M1 (structure =) simple / molecular, because it has a low melting / boiling point M2 (bonding =) covalent, because it is hydrolysed	2
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56. O/N 19/P21/Q2/a,b(ii,iii),c(ii,iii)

2(a)	Na₂O	MgO	Al₂O₃	SiO₂	SO₃	2
	ionic	ionic	ionic	covalent	covalent	
	giant	giant	giant	giant / macro-molecular	simple / molecular	
Award one mark for each correct row.						

2(b)(ii)	M1: reacts with both acid and base / alkali M2: use any equation with Al₂O₃ and an acid, e.g. Al₂O₃ + 6HCl → 2AlCl₃ + 3H₂O M3: use any equation with Al₂O₃ and a base / alkali, e.g. Al₂O₃ + 2NaOH + 3H₂O → 2NaAl(OH)₄	3
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2(b)(iii)	solid dissolves / disappears OR gets warm / hot	1
2(c)(ii)	M1: use of the correct expression in terms of specific bond energies. ($514 - xE_{\text{Se}-\text{O}} = -346$) M2: use of correct stoichiometry AND correct processing of expression given in M1. Provided the values 514 and 346 are used. ($514 - 2E_{\text{Se}-\text{O}} = -346$) = (+)430 (kJ mol ⁻¹)	2
2(c)(iii)	$\text{SeO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SeO}_3 + \text{H}_2\text{O}$	1

57. O/N 19/P22/Q2/a,b(i)

2(a)	<table><tr><td>Na₂O</td><td>Al₂O₃</td><td>SiO₂</td><td>P₄O₁₀/P₄O₆/P₂O₃/P₂O₅</td><td>SO₃</td></tr><tr><td>basic</td><td>amphoteric</td><td>acidic</td><td>acidic</td><td>acidic</td></tr></table>					Na ₂ O	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀ /P ₄ O ₆ /P ₂ O ₃ /P ₂ O ₅	SO ₃	basic	amphoteric	acidic	acidic	acidic	2
	Na ₂ O	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀ /P ₄ O ₆ /P ₂ O ₃ /P ₂ O ₅	SO ₃											
	basic	amphoteric	acidic	acidic	acidic											
	M1: all formulae correct															
M2: all acid / base behaviour correctly stated																
2(b)(i)	reacts with both acid and base					1										

58. O/N 19/P22/Q3/a(i,ii)

3(a)(i)	green gas fades OR white solid / white powder / white smoke / white fumes	1
3(a)(ii)	hydrolysis	1

59. O/N 19/P23/Q2/a,b(ii,iii),c(ii,iii)

2(a)	<table><tr><td>Na₂O</td><td>MgO</td><td>Al₂O₃</td><td>SiO₂</td><td>SO₃</td></tr><tr><td>ionic</td><td>ionic</td><td>ionic</td><td>covalent</td><td>covalent</td></tr><tr><td>giant</td><td>giant</td><td>giant</td><td>giant / macro-molecular</td><td>simple / molecular</td></tr></table> Award one mark for each correct row.	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	SO ₃	ionic	ionic	ionic	covalent	covalent	giant	giant	giant	giant / macro-molecular	simple / molecular	2
Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	SO ₃													
ionic	ionic	ionic	covalent	covalent													
giant	giant	giant	giant / macro-molecular	simple / molecular													
2(b)(ii)	M1: reacts with both acid and base / alkali M2: use any equation with Al₂O₃ and an acid, e.g. Al₂O₃ + 6HCl → 2AlCl₃ + 3H₂O M3: use any equation with Al₂O₃ and a base / alkali, e.g. Al₂O₃ + 2NaOH + 3H₂O → 2NaAl(OH)₄	3															
2(b)(iii)	solid dissolves / disappears OR gets warm / hot	1															
2(c)(ii)	M1: use of the correct expression in terms of specific bond energies. (514 – xE_{Se–O} = –346) M2: use of correct stoichiometry AND correct processing of expression given in M1. Provided the values 514 and 346 are used. (514 – 2E_{Se–O} = –346) = (+)430 (kJ mol^{–1})	2															
2(c)(iii)	SeO ₂ + 2NaOH → Na ₂ SeO ₃ + H ₂ O	1															

60. M/J 19/P21/Q1/b

1(b)(i)	Similarities (any two from the following list) (both have) +2 ion / (+2) same oxidation state / same stoichiometry of oxide / carbonates decompose (on heating)	2
	Difference (X) forms coloured compounds/oxides/ carbonates OR Group 2 elements form white compounds/oxides/carbonates	1
1(b)(ii)	XO	1
1(b)(iii)	$\text{XCO}_3 \rightarrow \text{XO} + \text{CO}_2$	1

61. M/J 19/P21/Q2/b,f

2(b)	$\text{Mg}_2\text{Si}(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Mg}(\text{OH})_2(\text{aq}) + \text{SiH}_4(\text{g})$ M1 correct balancing and formulae	1
	M2 state symbols	1
2(f)(i)	M1 sodium silicate / Na_2SiO_3	1
	M2 water / H_2O	1
2(f)(ii)	acid(ic)	1

62. M/J 19/P22/Q3/a(i,ii,iii)

3(a)(i)	$\text{SiCl}_4(\text{l}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{SiO}_2(\text{s}) + 4\text{HCl}(\text{aq} / \text{g})$ (state symbols required)	1
3(a)(ii)	hydrolysis	1
3(a)(iii)	NaCl – ionic	1
	SiCl_4 – covalent	1

63. MAR 19/P22/Q1/b(ii)

1(b)(ii)	solid disappears	1
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64. O/N 18/P22/Q2/a,c,d

2(a)	<table><tr><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>metallic</td><td>metallic</td><td>metallic</td><td>covalent</td><td>covalent</td></tr><tr><td>Na₂O</td><td>MgO</td><td>Al₂O₃</td><td>SiO₂</td><td>P₄O₁₀</td></tr><tr><td>NaCl</td><td>MgCl₂</td><td>AlCl₃</td><td>SiCl₄</td><td>PCl₅</td></tr></table> [1] for each correct row	Na	Mg	Al	Si	P	metallic	metallic	metallic	covalent	covalent	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	NaCl	MgCl ₂	AlCl ₃	SiCl ₄	PCl ₅	3
Na	Mg	Al	Si	P																		
metallic	metallic	metallic	covalent	covalent																		
Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀																		
NaCl	MgCl ₂	AlCl ₃	SiCl ₄	PCl ₅																		
2(c)(i)	graph rises to maximum for Si, then falls	1																				
2(c)(ii)	Max. 3 from: - increasing strength of metallic bond; Na < Mg < Al OR stronger attraction between delocalised electrons and (positive) ion so melting point of Na < Mg < Al - Si is giant covalent AND it has the highest melting point due to breaking / presence of strong (covalent) bonds OR Si requires the most energy because the covalent bonds in Si are stronger than metallic bonds (in Na / Mg / Al) - P and S have weak(er) intermolecular forces / induced dipoles / van der Waals forces (than covalent / metallic bonds) so have low(er) melting points - S₍₈₎ has stronger / more intermolecular forces / van der Waals forces / induced dipoles than P₍₄₎ so melting point of S₍₈₎ is higher	3																				
2(d)	P ₄ O ₁₀ + 6H ₂ O → 4H ₃ PO ₄	1																				

65. M/J 18/P21/Q3/a(i),c

3(a)(i)	increasing attraction between nucleus and (outer) electrons			1
	increasing nuclear charge with similar shielding / (electrons in) same (outer) shell			1
3(c)(i)	oxidation numbers / states of elements (Na-Si) increase from +1 to +4 / by 1 every time			1
	increasing number of valence electrons / NaCl , MgCl_2 , AlCl_3 , SiCl_4 / number of chlorines matches group number			1
	chlorine oxidation number / state -1 in all / stays the same			1
3(c)(ii)	$\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$			1
	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$			1
3(c)(iii)		structure	bonding	2
	sodium chloride	giant / ionic	ionic	
	silicon(IV) chloride	simple / molecular	covalent	

66. MAR 18/P22/Q1/b

1(b)	line on graph decreases P—T increasing nuclear charge AND electrons in same shell greater attraction between nucleus (and electrons)	3
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67. MAR 18/P22/Q2/d(iii,v)

2(d)(iii)	$\text{Mg}_2\text{Si(s)} + 4\text{HCl(aq)} \rightarrow \text{SiH}_4\text{(g)} + 2\text{MgCl}_2\text{(aq)}$ species AND balancing state symbols	2
2(d)(v)	$\text{SiH}_4 + 2\text{O}_2 \rightarrow \text{SiO}_2 + 2\text{H}_2\text{O}$	1

68. O/N 17/P21/Q2/b

2(b)(i)	(L=) MgCl_2 / magnesium chloride	1
	Any two from (giant) ionic (with strong attractions) $\text{Mg}^{2+}\text{(aq)} / \text{Mg}(\text{H}_2\text{O})_6^{2+}\text{(aq)}$ is neutral / undergoes (partial) hydrolysis $\text{Mg}(\text{OH})_2$ is the white precipitate / solid / insoluble / partially soluble $\text{MgCl}_2 + 2\text{NaOH} \rightarrow \text{Mg}(\text{OH})_2 + 2\text{NaCl}$	2
2(b)(ii)	(M=) SiCl_4 / silicon chloride	1
	Any two from (simple) molecular / simple covalent hydrolysis possible due to available d orbitals forms HCl(aq) / hydrochloric acid / solution and / or HCl(g) gas / fumes white solid is (hydrated) SiO_2 $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	2

69. O/N 17/P23/Q2/b

2(b)(i)	(L=) MgCl_2 / magnesium chloride	1
	Any two from (giant) ionic (with strong attractions) $\text{Mg}^{2+}\text{(aq)} / \text{Mg}(\text{H}_2\text{O})_6^{2+}\text{(aq)}$ is neutral / undergoes (partial) hydrolysis $\text{Mg}(\text{OH})_2$ is the white precipitate / solid / insoluble / partially soluble $\text{MgCl}_2 + 2\text{NaOH} \rightarrow \text{Mg}(\text{OH})_2 + 2\text{NaCl}$	2
2(b)(ii)	(M=) SiCl_4 / silicon chloride	1
	Any two from (simple) molecular / simple covalent hydrolysis possible due to available d orbitals forms HCl(aq) / hydrochloric acid / solution and / or HCl(g) gas / fumes white solid is (hydrated) SiO_2 $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	2

70. M/J 17/P21/Q3/a(i)

3(a)(i)	A	1
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71. MAR 17/P22/Q1/a(i,ii),b

1(a)(i)	<table><tr><td>max O.N.</td><td>+1</td><td>(+)2</td><td>(+)3</td><td>(+)5</td><td>(+)6</td><td>+7</td></tr></table>	max O.N.	+1	(+)2	(+)3	(+)5	(+)6	+7	1
max O.N.	+1	(+)2	(+)3	(+)5	(+)6	+7			
1(a)(ii)	(from Na to Cl) nuclear charge increases	1							
	electrons are in the same shell / have same shielding	1							
	greater / stronger attraction (of electrons to nucleus)	1							
1(b)(i)	$P_4 + 5O_2 \rightarrow P_4O_{10} / 2P_2O_5$	1							
1(b)(ii)	any 2 from: • yellow / green colour (of chlorine gas) disappears • white flame • white solid • solid melts	2							
1(b)(iii)	phosphoric(V) acid	1							

72. O/N 16/P22/Q3/b

3(b)(i)	Increasing strength of / more energy needed to break (metallic) bonding / increasing strength of attraction between (cat)ion / nucleus and delocalised / free / sea of / cloud of electrons	1
	Increasing number of delocalised electrons / decreasing (cat)ion size / increasing charge / charge density of (cat)ion	1
3(b)(ii)	Attraction for electrons too strong to fully delocalise all 3 in Al OR difference in size between 12/Mg ²⁺ and 13/Al ³⁺ is less than difference in size between 11/Na ⁺ and 12/Mg ²⁺ OR magnitude of increase in charge is less from 2+ to 3+ than from 1+ to 2+	1
3(b)(iii)	Increase (15/P to 16/S) then decrease (to 17/Cl and 18/Ar)	1
	OR general decrease (from 15/P to 18/Ar) with an increase from 15/P to 16/S OR S ₍₈₎ > P ₍₄₎ > Cl ₍₂₎ > Ar	1
	(melting point depends on strength of) VdW / IMFs	1
	The greater the number of electrons in the molecule (atom for Ar) the greater the strength of VdW / IMFs OR the greater the melting point ora	1
3(b)(iv)	Giant covalent (structure) / many (strong) covalent bonds (need breaking)	1

73. M/J 16/P21/Q2/a,b,c,d

2 (a)	D = Ga G = Se	[1]	[1]
(b) (i)	$D_2O_3 + 6HCl \rightarrow 2DCl_3 + 3H_2O$ M1 = species; M2 = balancing	[1] [1]	[2]

(ii)	$\text{D}_2\text{O}_3 + 2\text{NaOH} + 7\text{H}_2\text{O} \rightarrow 2\text{NaD}(\text{OH})_4(\text{H}_2\text{O})_2$ OR $\text{D}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaD}(\text{OH})_4$ OR $\text{D}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaDO}_2 + \text{H}_2\text{O}$ OR $\text{D}_2\text{O}_3 + 2\text{OH}^- + 7\text{H}_2\text{O} \rightarrow 2[\text{D}(\text{OH})_4(\text{H}_2\text{O})_2]^-$ OR $\text{D}_2\text{O}_3 + 2\text{OH}^- + 3\text{H}_2\text{O} \rightarrow 2[\text{D}(\text{OH})_4]^-$ OR $\text{D}_2\text{O}_3 + 2\text{OH}^- \rightarrow 2\text{DO}_2^- + \text{H}_2\text{O}$ M1 = species; M2 = balancing	[1] [1]	[2]
(c)	giant ionic / ionic lattice	[1]	[1]
(d)	$\text{GO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{GO}_3$	[1]	[1]

74. M/J 16/P22/Q2/d

(d) (i)	no of <i>Cl</i> increases <u>by one</u> each time / matches group number	[1]	
	due to increasing number of valence/outer(most/shell) electrons / oxidation number / valency (of Mg, Al, Si)	[1]	[2]
(ii)	$\text{MgCl}_2 (+\text{aq}) \rightarrow \text{Mg}^{2+} + 2\text{Cl}^-$ $\text{AlCl}_3 + 6\text{H}_2\text{O} \rightarrow \text{Al}(\text{H}_2\text{O})_6^{3+} + 3\text{Cl}^-$ / $\text{Al}(\text{H}_2\text{O})_5(\text{OH})^{2+} + \text{H}^+ + 3\text{Cl}^-$ $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{H}^+ + 4\text{Cl}^-$	[1] [1] [1]	[3]

75. M/J 16/P23/Q1/b(ii)

(ii)	$\text{FeCl}_3 + 6\text{H}_2\text{O} \rightarrow [\text{Fe}(\text{H}_2\text{O})_6]^{3+} 3\text{Cl}^-$ OR $\text{FeCl}_3 + 6\text{H}_2\text{O} \rightarrow [\text{Fe}(\text{H}_2\text{O})_6\text{OH}]^{2+} + \text{H}^+ + 3\text{Cl}^-$	[1]	[1]
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76. MAR 16/P22/Q1/a(i,iii)

1 (a) (i)	greater <u>attractive</u> force OR greater force <u>between nucleus and (outer) electrons</u> proton number / atomic number / nuclear charge increases across period AND electrons occupy same shell / shielding roughly constant	[1] [1]	[2]
(iii)	sodium has mobile / free electrons / electrons free (to move throughout the structure) phosphorus is simple / covalent / molecular	[1] [1]	[2]

77. O/N 15/P22/Q3/a,b

3 (a) (i)	Na_2O or Na_2O_2 ; MgO ; P_4O_{10} or P_4O_6 ; SO_2	[1] [1]	[2]
(ii)	Na: Yellow / orange / gold flame / white solid / powder / smoke $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$ or $2\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}_2$ S: Blue flame / (yellow) solid melts / turns red / amber / white fumes $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$	[1] [1] [1] [1]	[4]
(b) (i)	acidic P and S amphoteric Al and basic Na and Mg	[1] [1]	[2]
(ii)	acidic: covalent (bonding) basic: ionic (bonding)	[1] [1]	[2]
(iii)	$\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ OR $\text{Al}_2\text{O}_3 + 6\text{H}^+ \rightarrow 2\text{Al}^{3+} + 3\text{H}_2\text{O}$ $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 7\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4(\text{H}_2\text{O})_2$ OR $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ OR $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$ OR $\text{Al}_2\text{O}_3 + 2\text{OH}^- + 7\text{H}_2\text{O} \rightarrow 2[\text{Al}(\text{OH})_4(\text{H}_2\text{O})_2]^-$ OR $\text{Al}_2\text{O}_3 + 2\text{OH}^- + 3\text{H}_2\text{O} \rightarrow 2[\text{Al}(\text{OH})_4]^-$ OR $\text{Al}_2\text{O}_3 + 2\text{OH}^- \rightarrow 2\text{AlO}_2^- + \text{H}_2\text{O}$	[1] [1]	[2]

78. M/J 15/P21/Q1/c(ii,iii),d

(c) (ii)	Covalent AND simple / molecular	[1]	[2]
	low melting point / reaction with water	[1]	
(iii)	$\text{TeCl}_4 + 3\text{H}_2\text{O} \rightarrow \text{H}_2\text{TeO}_3 + 4\text{HCl}$ OR $\text{TeCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{TeO}_2 + 4\text{HCl}$	[1]	[1]
(d) (i)	Yellow / orange flame	[1]	[max 2]
	White fumes / solid	[1]	
	Yellow / green gas disappears	[1]	
(ii)	NaCl giant / lattice AND ionic	[1]	[4]
	SiCl_4 simple / molecular AND covalent	[1]	
	For NaCl large difference in electronegativity (of sodium / Na and chlorine / Cl / Cl_2) (indicates electron transfer / ions)	[1]	
	For SiCl_4 smaller difference (indicates sharing / covalency) with (weak) van der Waals' / IM forces (between molecules) ora	[1]	

79. M/J 15/P22/Q1/d

(d) (i)	Na_2O basic / alkaline; Al_2O_3 amphoteric / acidic and basic; SO_3 acidic	[1]	[2]
	Na_2O (giant) ionic AND SO_3 (simple / molecular) covalent	[1]	
(ii)	$\text{Na}_2\text{O} + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O}$	[1]	[4]
	$\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$	[1]	
	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 7\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4(\text{H}_2\text{O})_2$ OR	[1]	
	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ OR		
	$\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$ OR		
	$\text{Al}_2\text{O}_3 + 2\text{OH}^- + 7\text{H}_2\text{O} \rightarrow 2[\text{Al}(\text{OH})_4(\text{H}_2\text{O})_2]^-$ OR		
	$\text{Al}_2\text{O}_3 + 2\text{OH}^- + 3\text{H}_2\text{O} \rightarrow 2[\text{Al}(\text{OH})_4]^-$ OR		
	$\text{Al}_2\text{O}_3 + 2\text{OH}^- \rightarrow 2\text{AlO}_2^- + \text{H}_2\text{O}$		
	$\text{SO}_3 + \text{NaOH} \rightarrow \text{NaHSO}_4$ OR	[1]	
	$\text{SO}_3 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$		